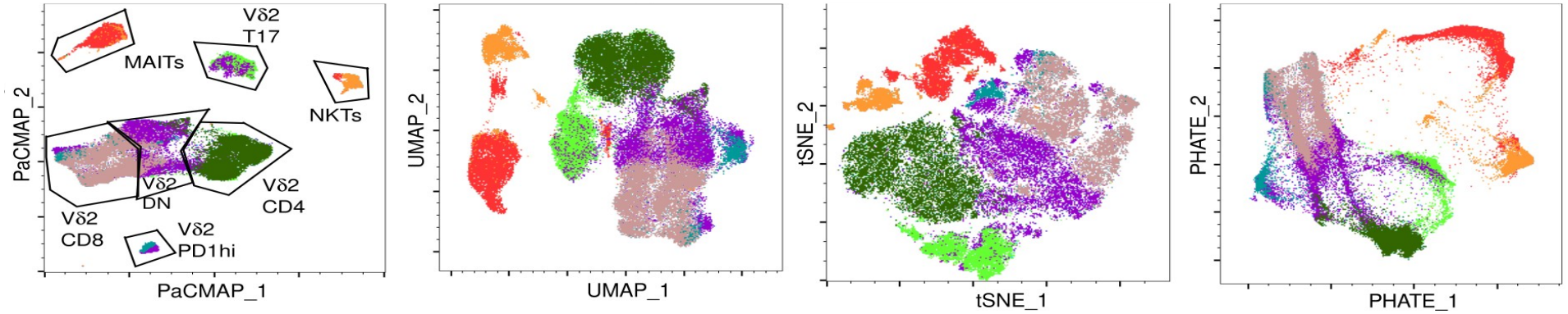
UNIVERSITY of MARYLAND
MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

ISAC RECOGNIZED
Shared Resource Laboratory




# Towards ‘No Cell Left Behind’: open-source tool sets to leverage Spectral Flow Cytometry’s full biological discovery potential

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UMGCCC Flow Cytometry Shared Resource
Baltimore, MD

October 23, 2026

Slides

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# Spectral Flow Cytometry is amazing!!!

- Capable of resolving remarkably similar fluorophores → **increased number of markers per panel** compared to conventional flow cytometry (CFC)
- Allows for rapid acquisition of millions of events in a few minutes → typically **greater number of acquired cells per sample** compared to mass cytometry (MC)
- Spectral Flow Cytometry (SFC) datasets have an unique combination of **width and depth** compared to preceding technologies






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# Spectral Flow Cytometry is amazing!!!



- Able to efficiently acquire greater number of cells → carry out a more thorough profiling of **rare cell subsets**
- Build one panel, evaluate multiple cell types → invaluable for clinical studies with **limited biospecimen**
- Expand our knowledge about immune system heterogeneity



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# Trade-off: Increasing analytical complexity

- If you are only interested in a **single population** of cells → analysis process remains <u>essentially unchanged</u> (manual gating with a few more gates)

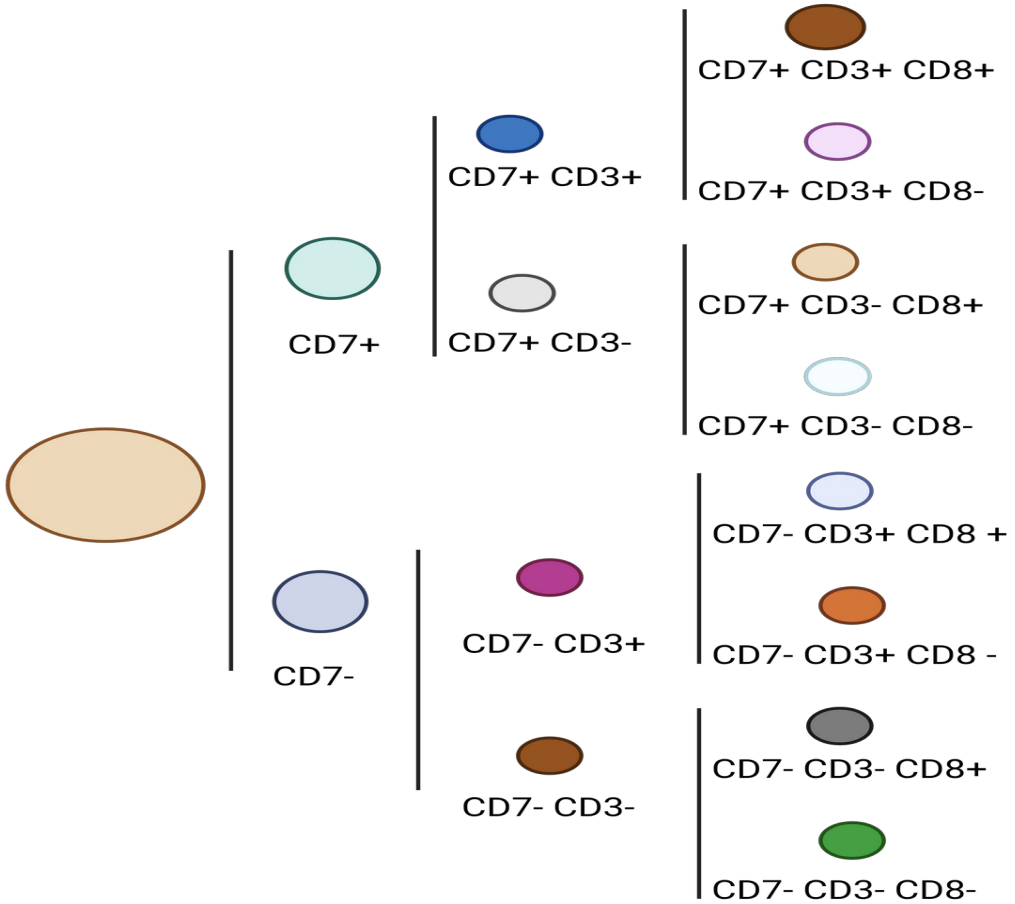
- If your goal is to characterize **all cell populations** that may be present → the breadth and depth of the datasets complicate the analysis






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# Complex Datasets: Combinatorially Exponential Clusters



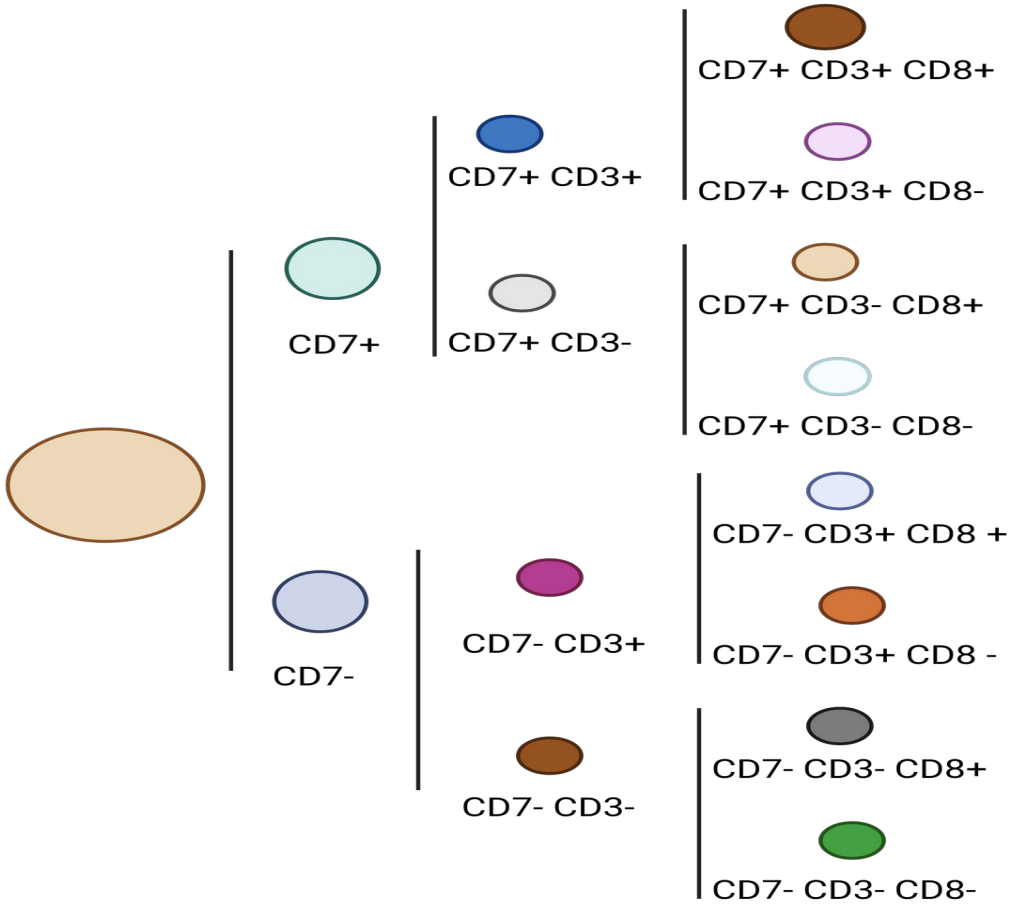

- If for a given marker in your panel, you **split** the cells into **positive** and **negative** populations
- And this splitting process is **repeated for every marker** in your panel
- At the end, we will have **grouped cells** that only **expressed the same markers**

- "Groups" per panel = $2^{\text{Number of Fluorophores}}$



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# Complex Datasets: Combinatorially Exponential Clusters




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- **9-color** CFC panel → **512** groups at most
- **40-color** SFC panel → **1.1e+12** groups (**trillion**)
- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)



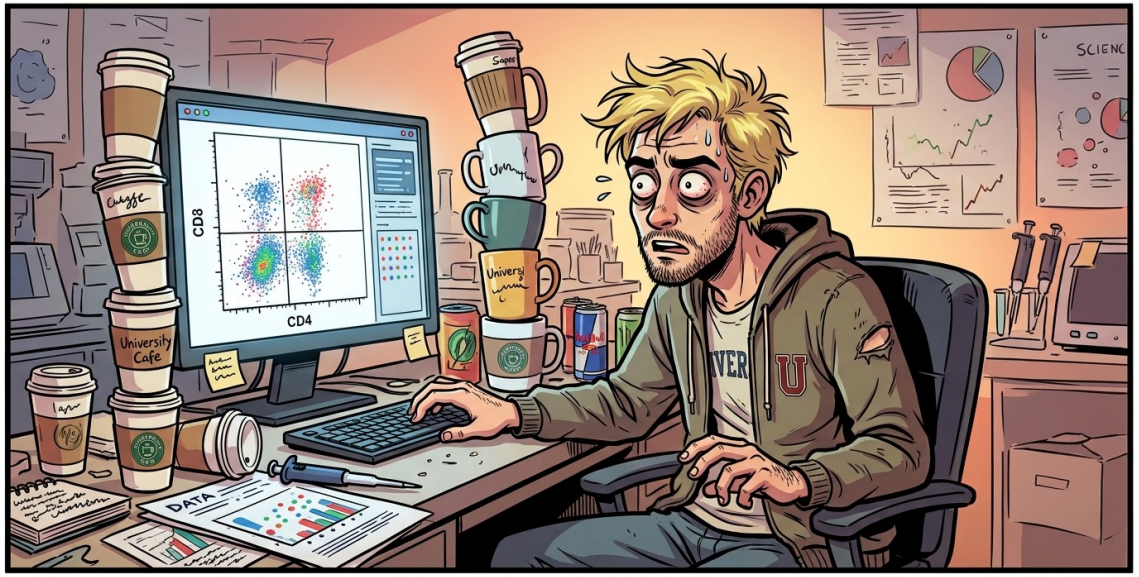
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# What even is a quintillion?

If I were to manually check for every combination...

- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 1 gate, every 10 seconds → 6 groups/minute
  - 60 minutes in an hour → 360 groups/hour
  - Awake 16 hours a day → 5760 groups/day
  - 365 days per year → **2,102,400 groups/year**






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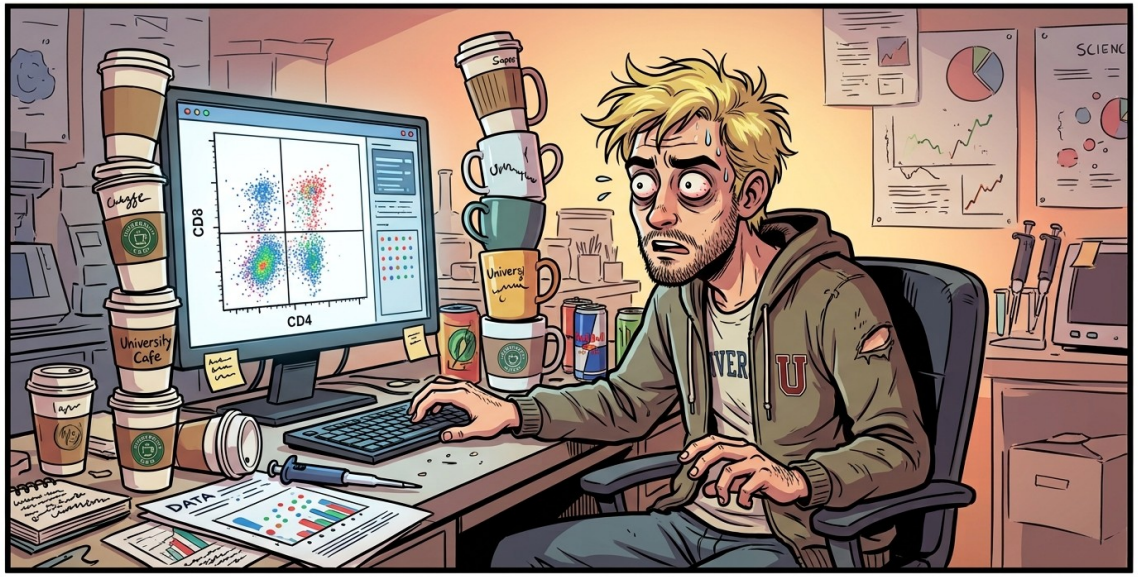
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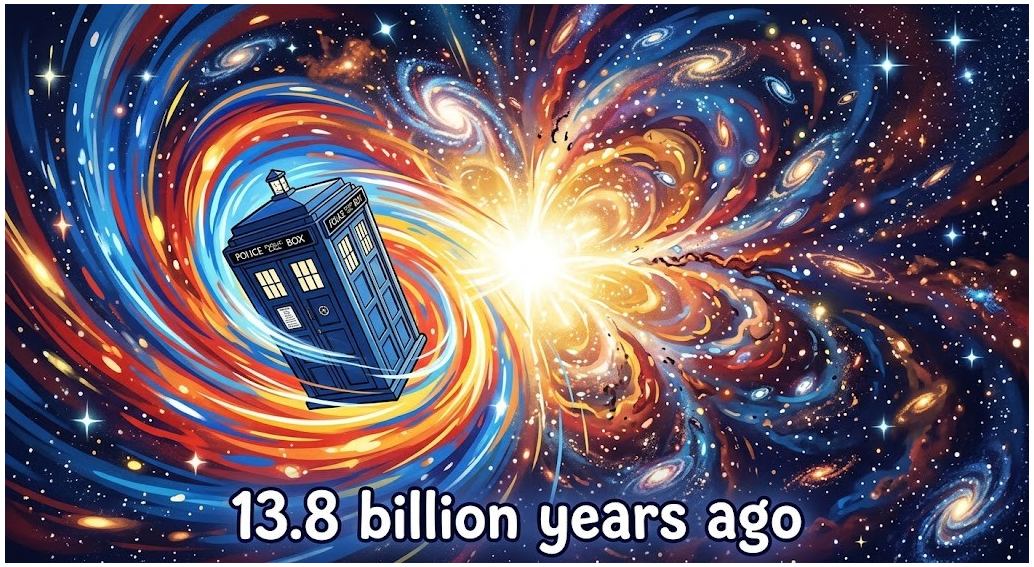
….Not even close….






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# Continuously since the Big Bang




- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**



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  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**



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# Continuously since the Big Bang




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  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**

1 file down, 99 more to go!



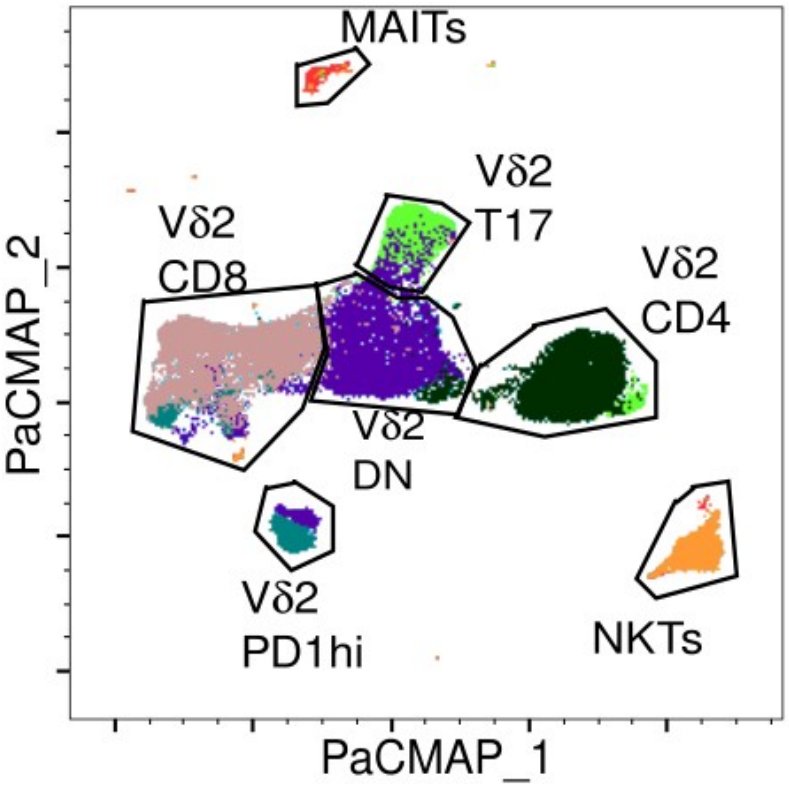
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# Alternative: Unsupervised Analysis (Algorithms)

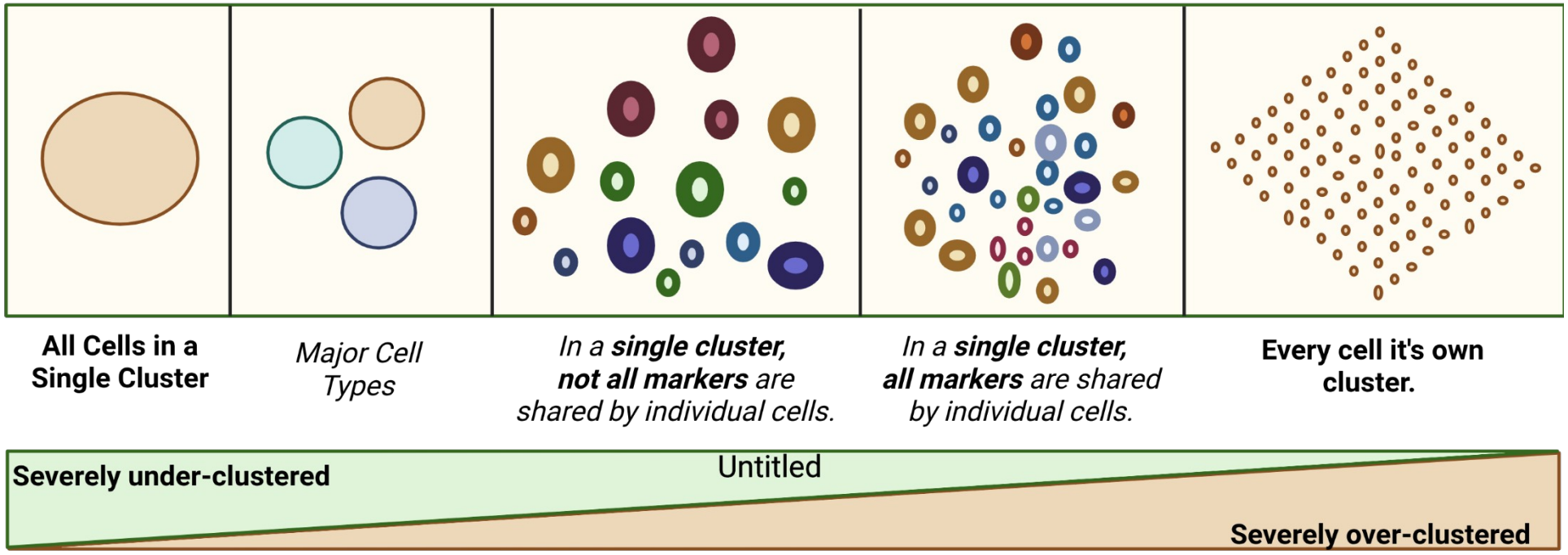
- Consequently, if the goal is to characterize all cell populations present, some form of **unsupervised (algorithm) or semi-supervised** analytical approaches will be needed

- **Clustering** → FlowSOM, Phenograph, Xshift, etc.
- **Dimensionality visualization** → opt-SNE, UMAP, PaCMAP, Phate, etc.

- **Automated Gating** → openCyto, flowMagic, ElasticGate, etc.



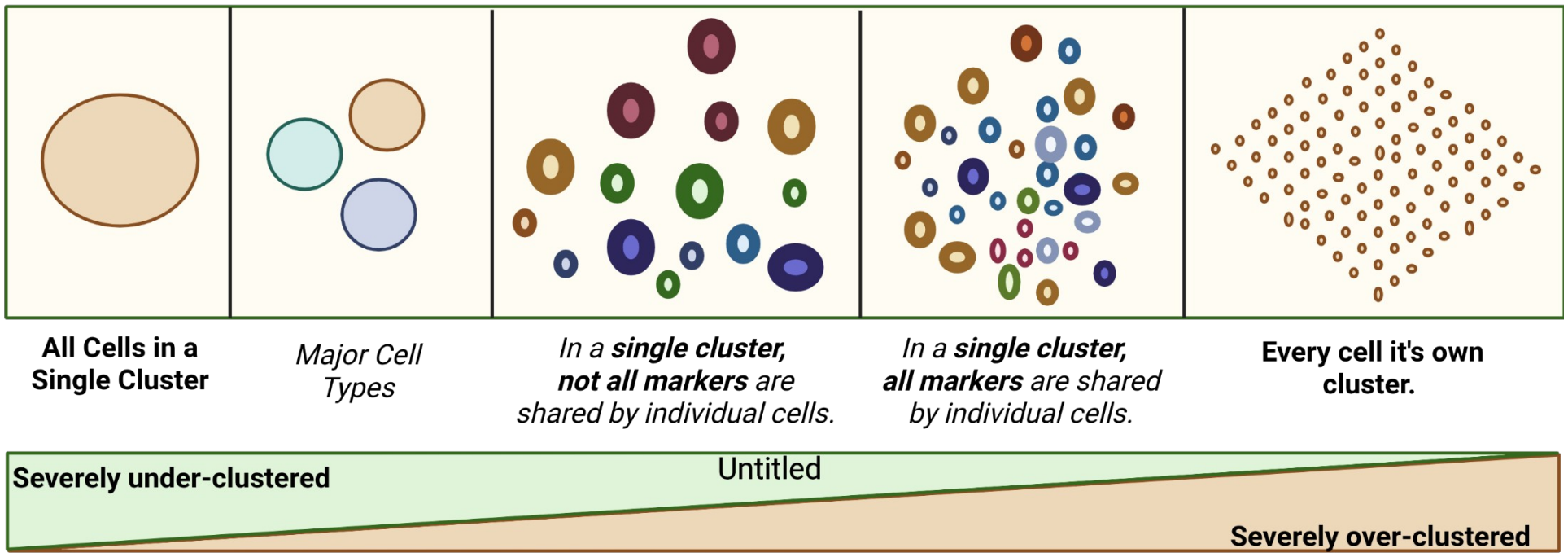
# The challenge for unsupervised approaches






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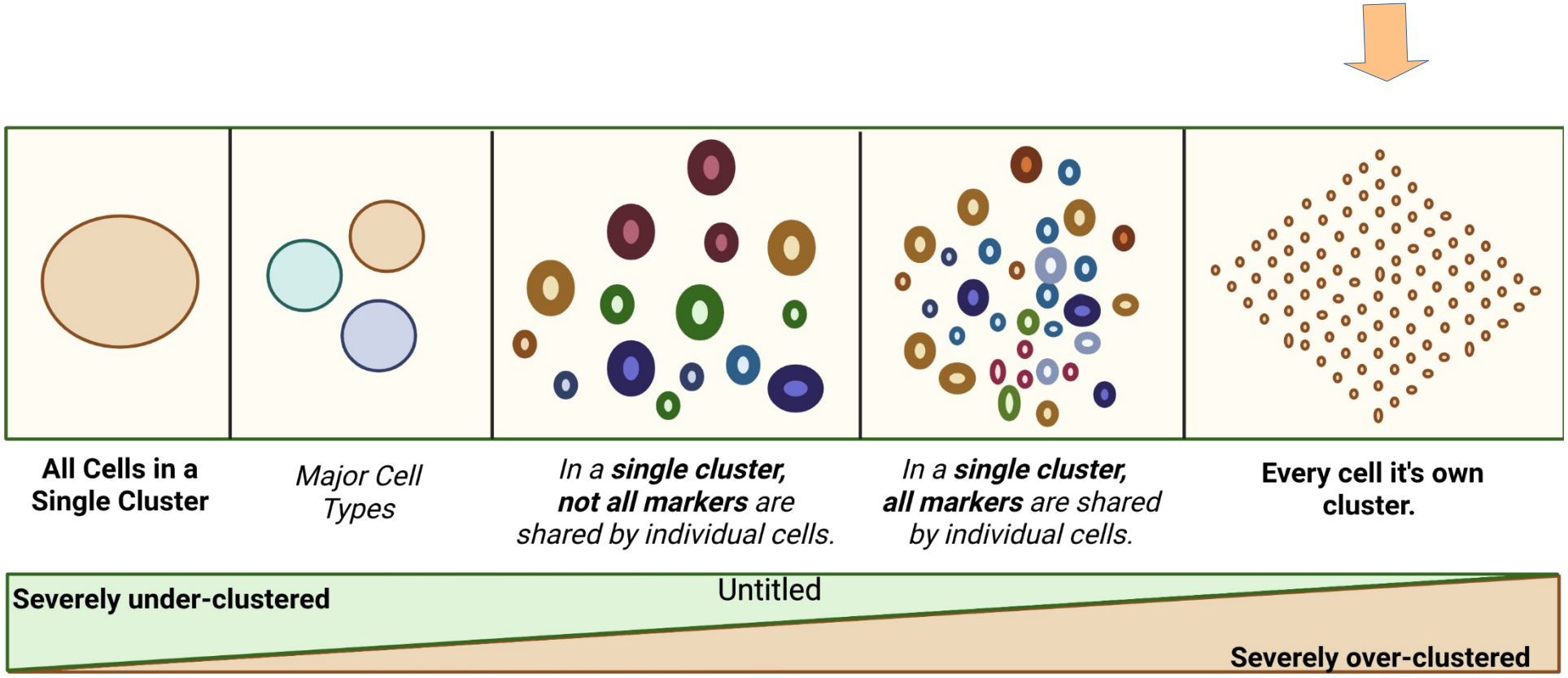
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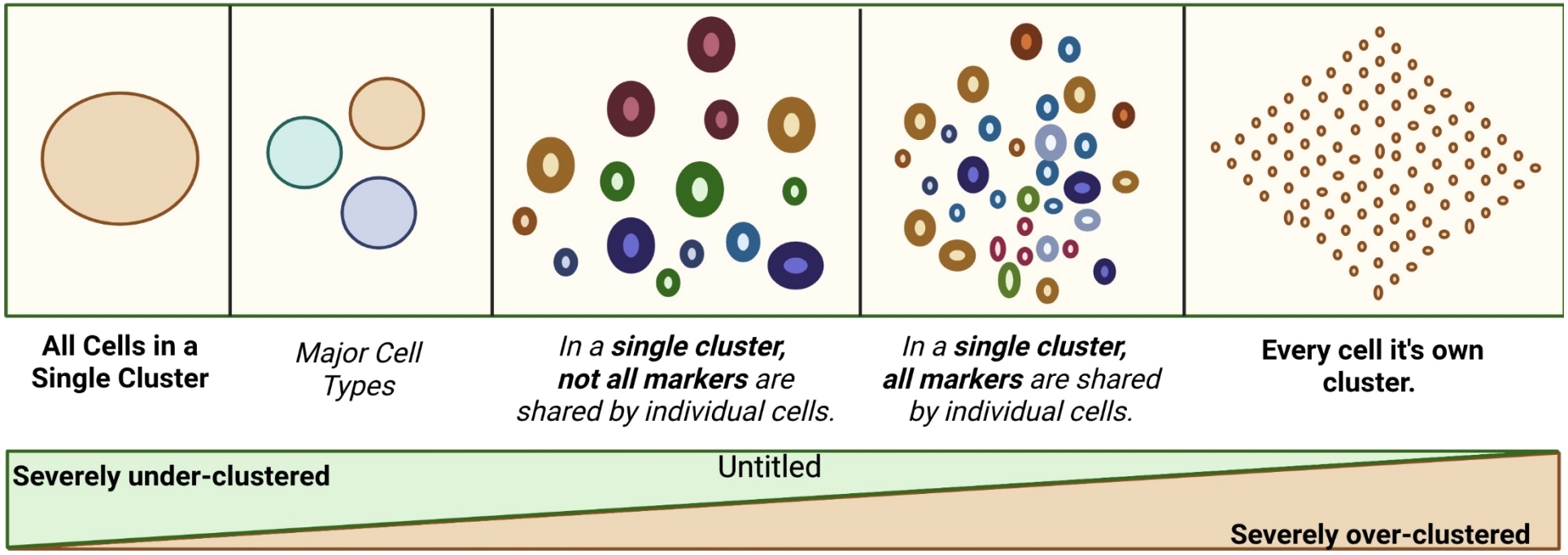
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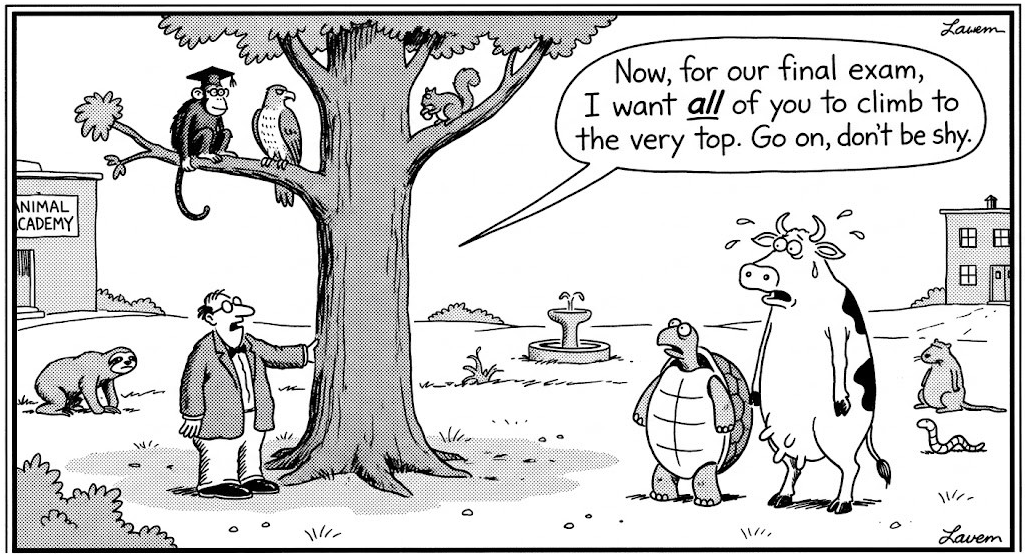
# The challenge for unsupervised approaches






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# Adapted algorithmic tools struggle with number of events




- Many of these software tools were originally implemented as R or Python packages, for use in analysis of **mass cytometry** or **single-cell RNA-seq datasets**
- Mass cytometry → similar number of markers, typically **fewer events**, no unmixing variation due to autofluorescence
- Single-cell RNA-seq → substantially more markers, typically way **fewer events.** No equivalent "negative" gate events for not expressed transcripts

- **Down-sampling becomes often necessary**



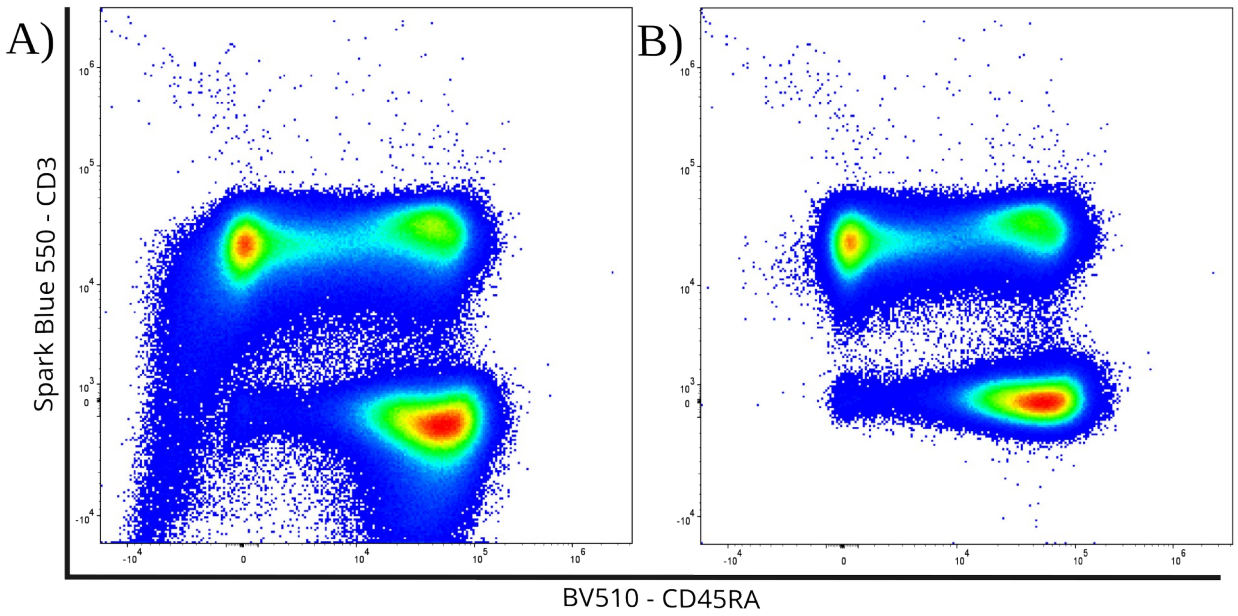
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# Adapted algorithmic tools struggle with unmixing errors

- Adapted software packages heavily rely on having a consistent **median fluorescent intensity (MFI)** to cluster cells based on positive or negative marker expression

- Existing **batch correction** / normalization / scaling tools target instrumental or **experimental** level variation for correction

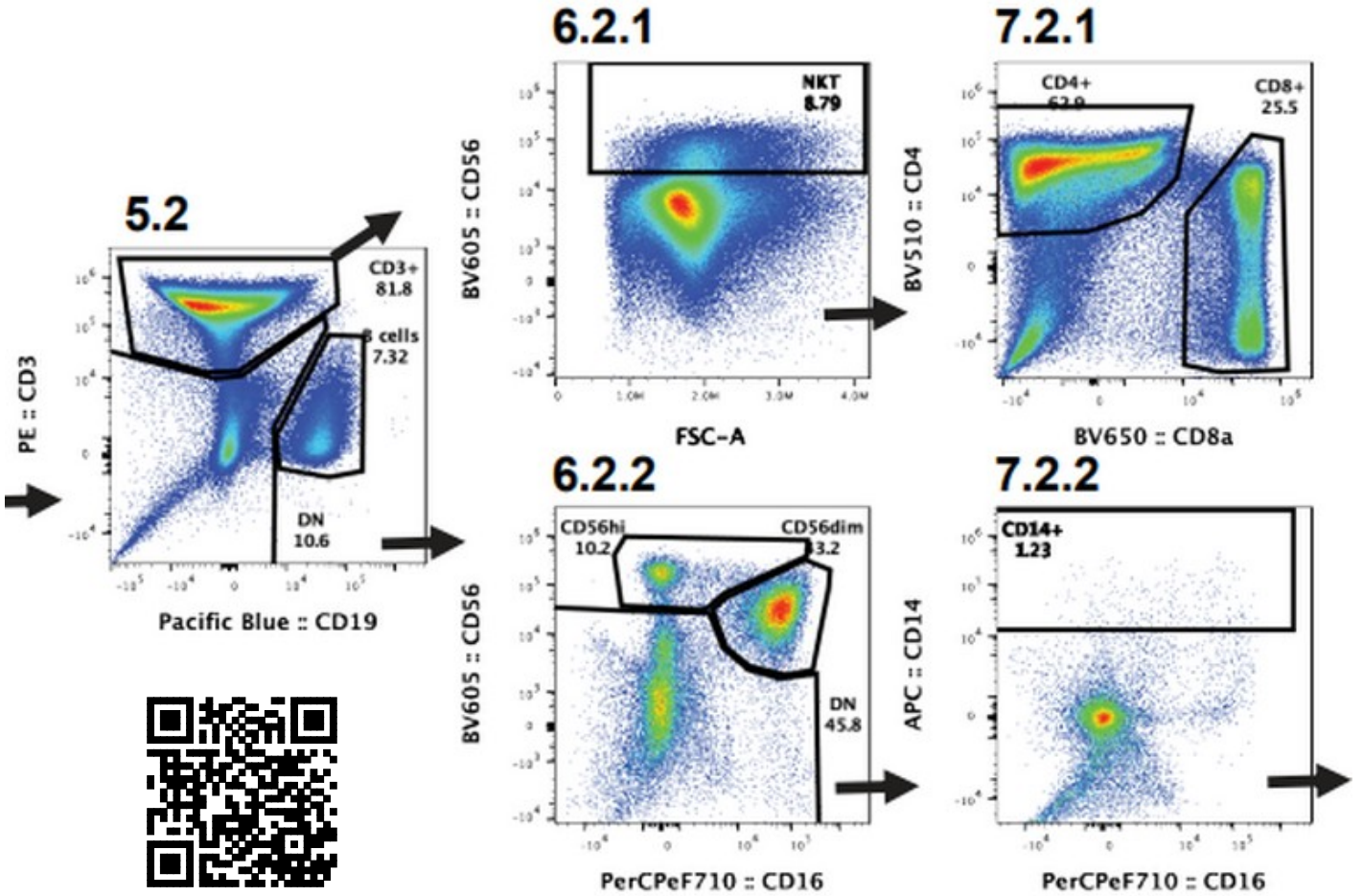
- **Unmixing errors** can seriously shift the MFI ranges for “negative” and “positive” cells for **individual** acquired specimens






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# “We discovered a novel dendritic cell cluster”






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# Full biological discovery potential often out of reach

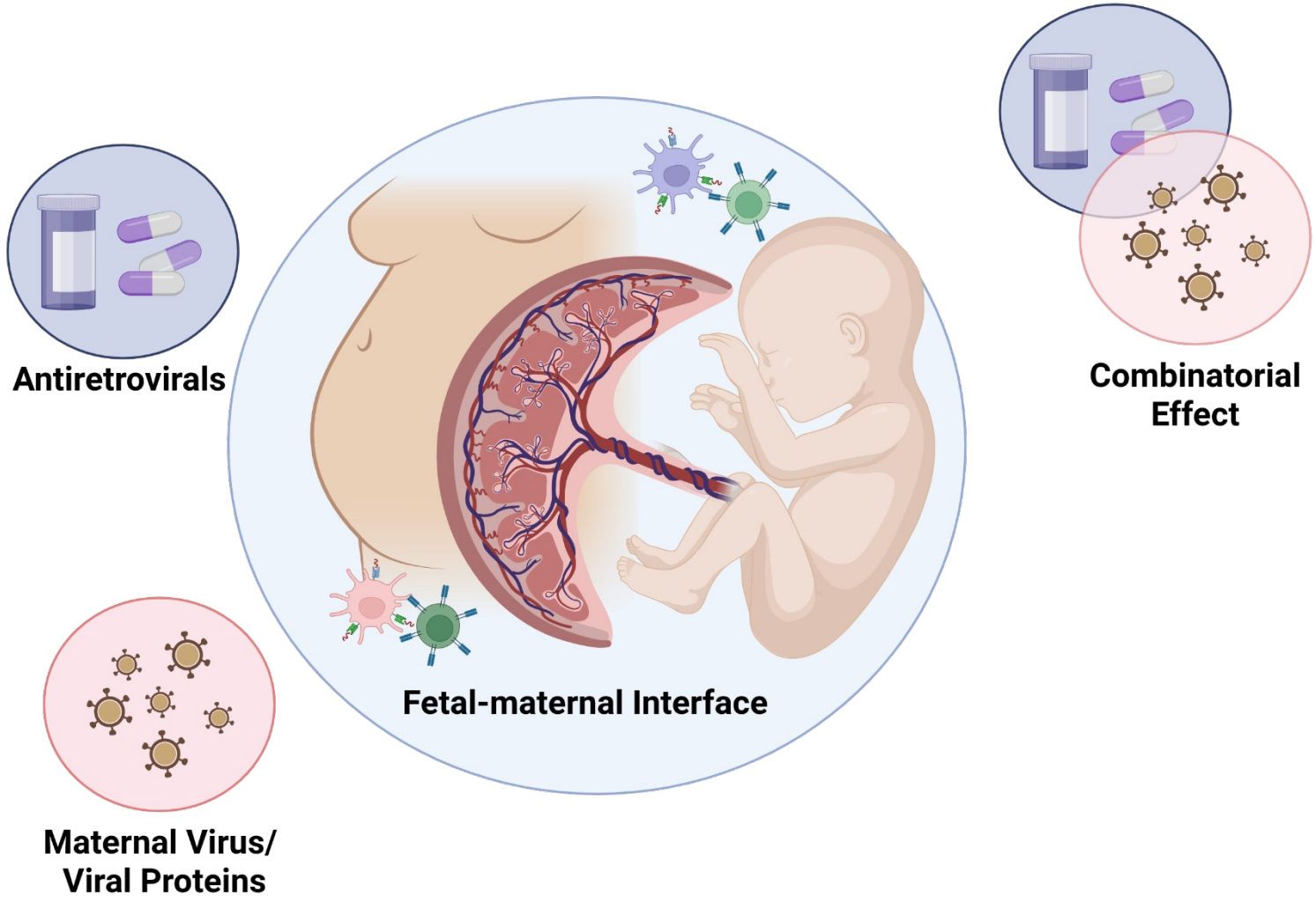



- For **most** spectral flow cytometry datasets, only a **handful of cell populations** end up being analyzed by any method



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# My PhD's focus: HIV-exposed Uninfected Infants

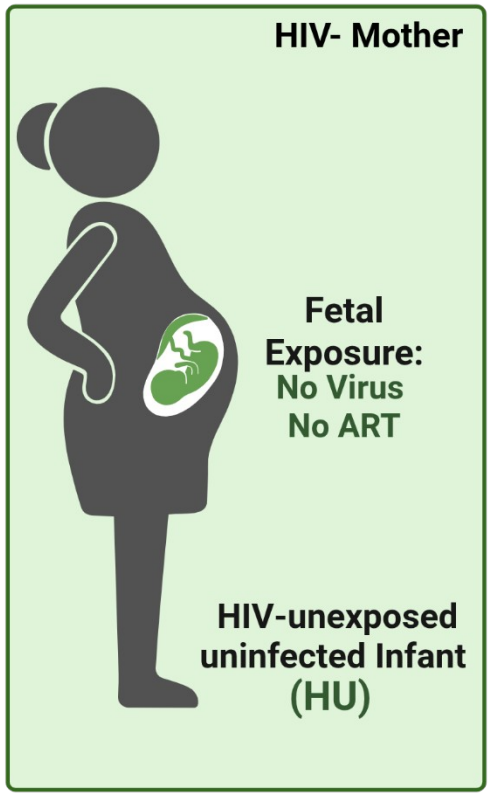


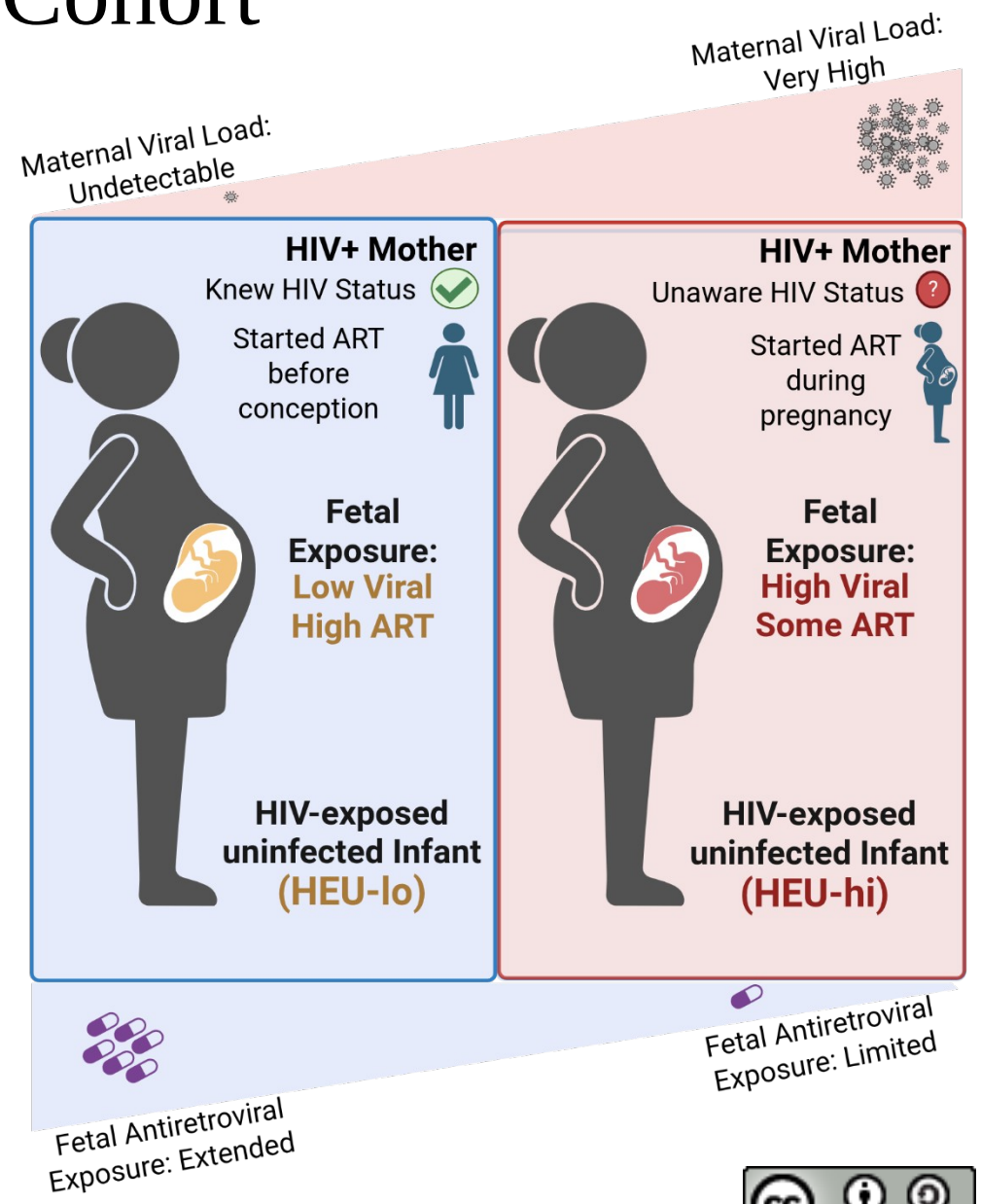



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# Our Malawian Cohort





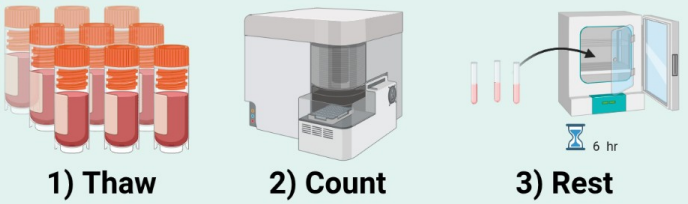

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# 32-Fluorophore Spectral Flow Cytometry Panel

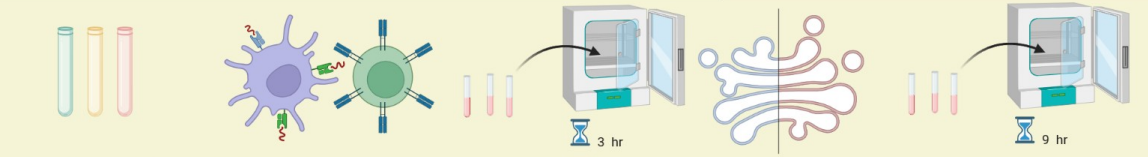
| | | | |
|---|---|---|---|
| UV1 | Spark UV 387 | | |
| UV2 | BUV395 | CD62L | (SK11) |
| UV6 | cFluor UV440 | | |
| UV7 | BUV496 | | |
| UV9 | BUV563 | CD69 | (FN50) |
| UV10 | BUV615 | CCR4 | (1G1) |
| UV11 | BUV661 | Vδ2 | (B6) |
| UV14 | BUV737 | CD38 | (HB7) |
| UV16 | BUV805 | CD4 | (SK3) |
| V1 | BV421/ | | |
| V2 | Spark Violet 423 | | |
| V3 | Pacific Blue | CD14 | (M5E2) |
| V3 | Pacific Blue | CD19 | (SJ25C) |
| V4 | SYTOX Blue | | |
| V5 | BV480 | CD161 | (HP-3G10) |
| V6 | SBV515 | | |
| V7 | BV510 | | |
| V8 | BV570 | CD16 | (3G8) |
| V9 | Qdot 585 | | |
| V10 | BV605 | CD45RA | (HI100) |
| V11 | BV650 | CD8 | (RPA-T8) |
| V13 | BV711 | Va7.2 | (3C10) |
| V14 | BV750 | IFNγ | (B27) |
| V15 | BV786 | CCR6 | (11A9) |
| V16 | LD NIR 786 | | |
| B2 | Alexa Fluor 488 | FoxP3 | (236A/E7) |
| B3 | Spark Blue 550 | CD3 | (SK7) |
| B4 | Spark Blue 574 | CD45 | (2D1) |
| B6 | RB613 | PD1 | (EH12.1) |
| B7 | NovaFluorBlue660 | | |
| B8 | SBB675 | | |
| B9 | NovaFluorBlue690 | | |
| B10 | RB705 | CD26 | (M-A261) |
| B12 | RB744/ | | |
| B13 | NovaFluorBlue760 | | |
| B14 | RB780 | CXCR5 | (RF8B2) |
| YG1 | PE | ICOS | (ISA-3) |
| YG2 | Spark YG 593 | | |
| YG3 | PE-Dazzle 594 | TNFα | (MAB11) |
| YG4 | PE-Fire 640/ | | |
| YG5 | PE-Cy5 | CXCR3 | (G025H7) |
| YG6 | NovaFluorYellow690 | | |
| YG7 | PE-Fire 700 | CD127 | (A019D5 ) |
| YG8 | PE-Fire 744 | CD25 | (M-A251) |
| YG9 | PE-Vio 770 | HLA-DR | (REA805) |
| YG10 | PE-Fire810 | | |
| R1 | APC | CD39 | (A1) |
| R2 | Alexa Fluor 647 | IL2 | (MQ1- |
| R3 | cFluor R685 | | |
| R4 | APC-R700 | CD107a | (H4A3) |
| R5 | NovaFluorRed725 | | |
| R6 | Zombie NIR | L/D | |
| R7 | APC-Fire 750 | CD27 | (O323) |
| R8 | APC-Fire 810 | CCR7 | (G043H7) |

**Thaw & Rest**



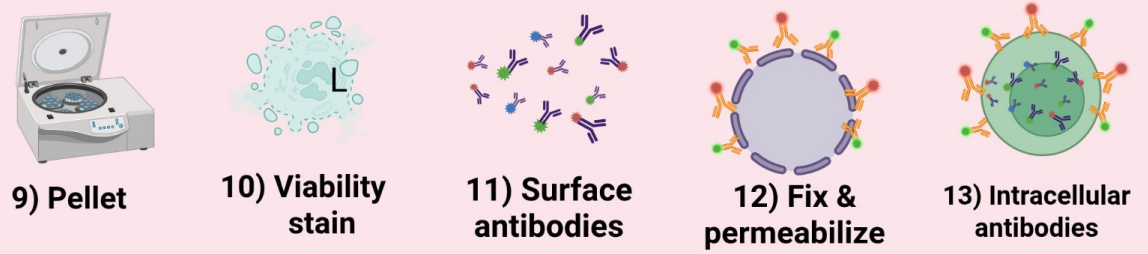

1) Thaw 2) Count 3) Rest

**Antigen Incubation**

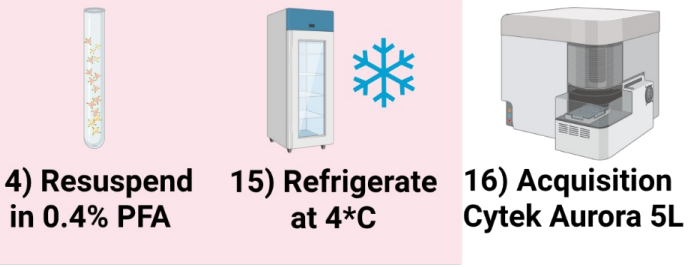



4) Aliquot by condition 5) Add antigens (PPD, SEB, Ctrl) 6) Incubate 7) Add Golgi Transport Inhibitors 8) Incubate

**Flow Cytometry Staining**




9) Pellet 10) Viability stain 11) Surface antibodies 12) Fix & permeabilize 13) Intracellular antibodies

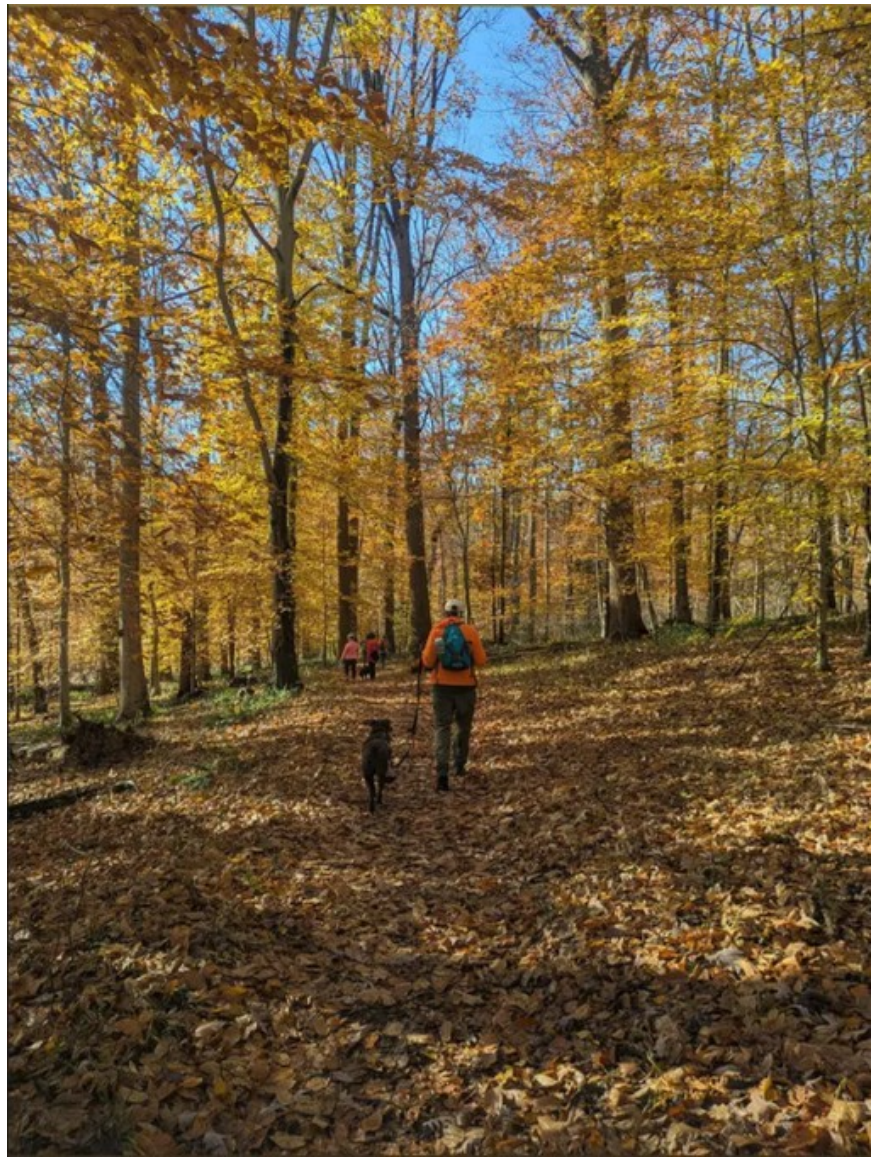


14) Resuspend in 0.4% PFA 15) Refrigerate at 4*C 16) Acquisition Cytek Aurora 5L

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# Personal Goal: No cell left behind



<u>My over-caffeinated at 3 AM thoughts:</u>

- ❑ What makes spectral flow cytometry data amazing is both the depth and the breadth

- ❑ How to fully profile the entirety of the acquired dataset, not just a single cell population of interest



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# Building open-source tool-sets to address SFC's quirks



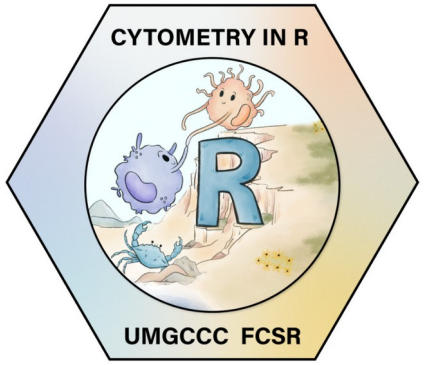

Spectral Signature Characterization and Isolation

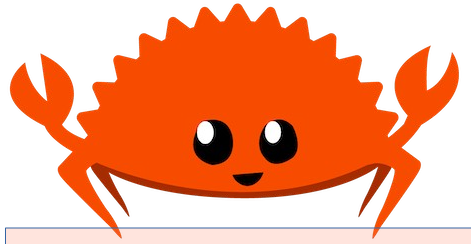



Incorporation of manually-annotated gating information




Automated Instrument Quality-Control Dashboards



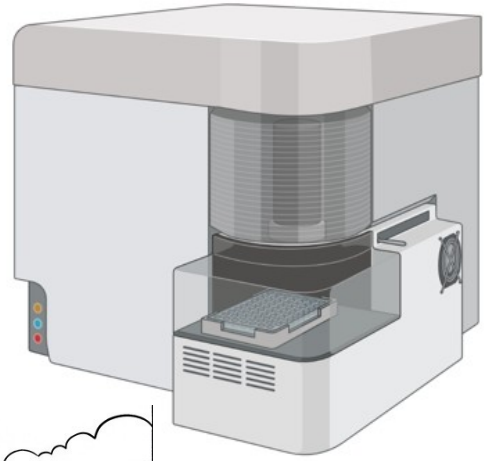



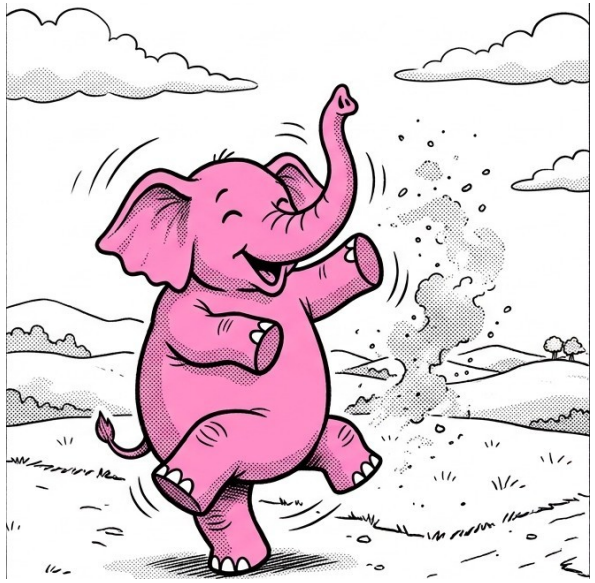
Rust



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# To unlock the comprehensive biological discovery potential, we need to address these pink elephants





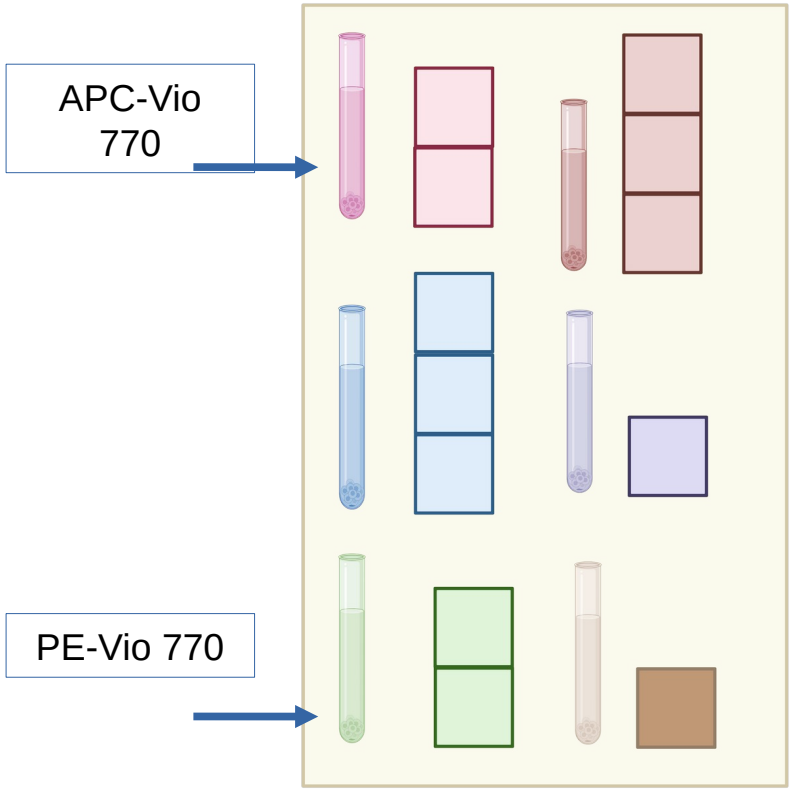
- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?

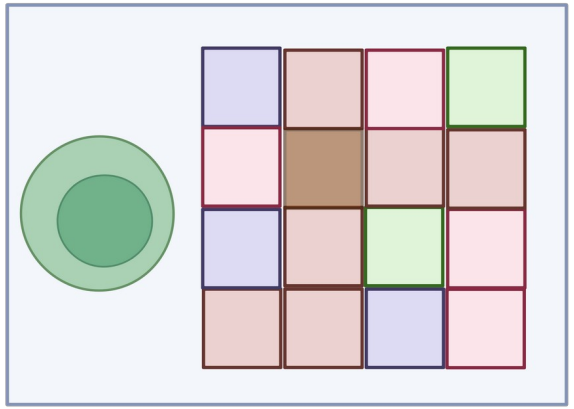


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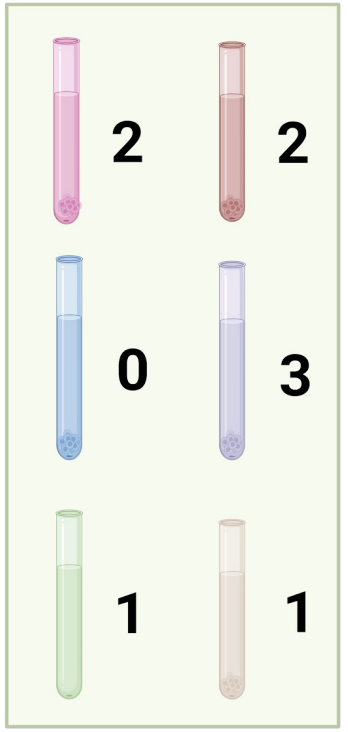
# Spectral Unmixing




Unmixing Controls (Single Color and Unstained)



Single cell from full-stained sample

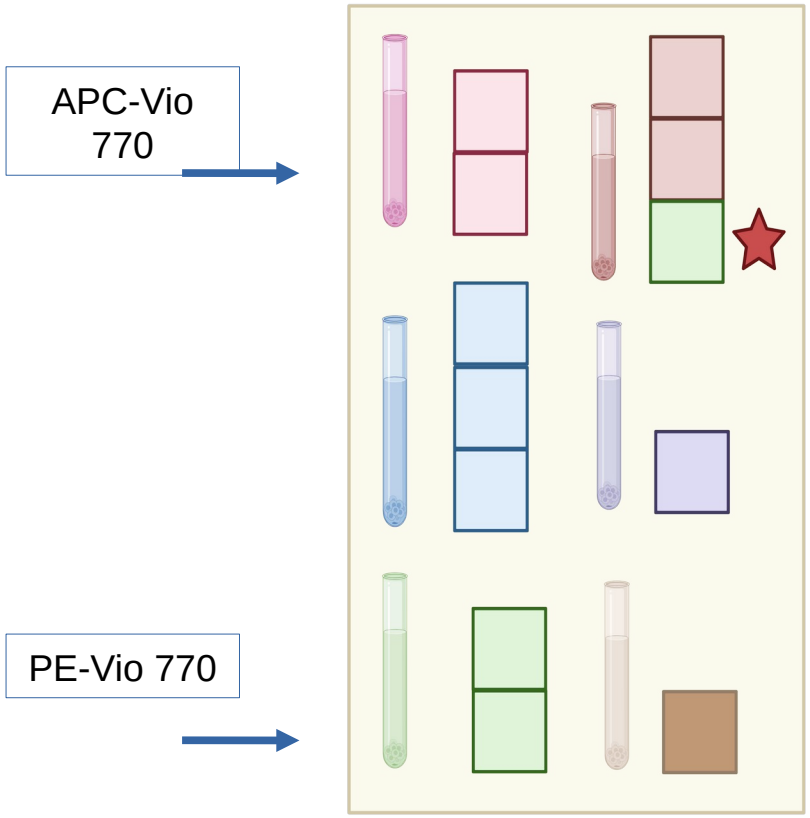



What fluorophores are present, and roughly how much

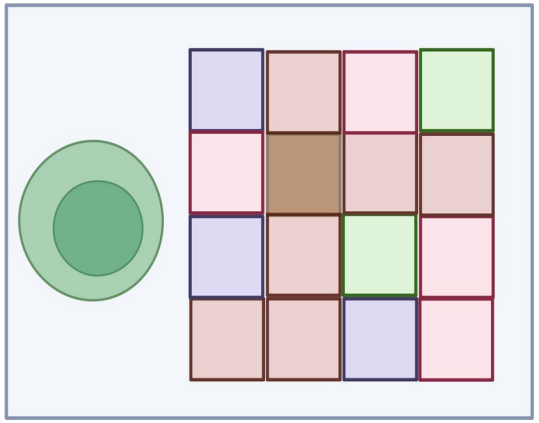


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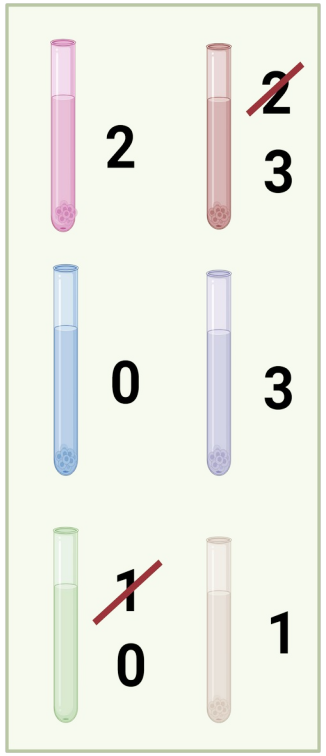
# Mismatched references lead to errors in attribution




Unmixing Controls (Single Color and Unstained)



Single cell from full-stained sample

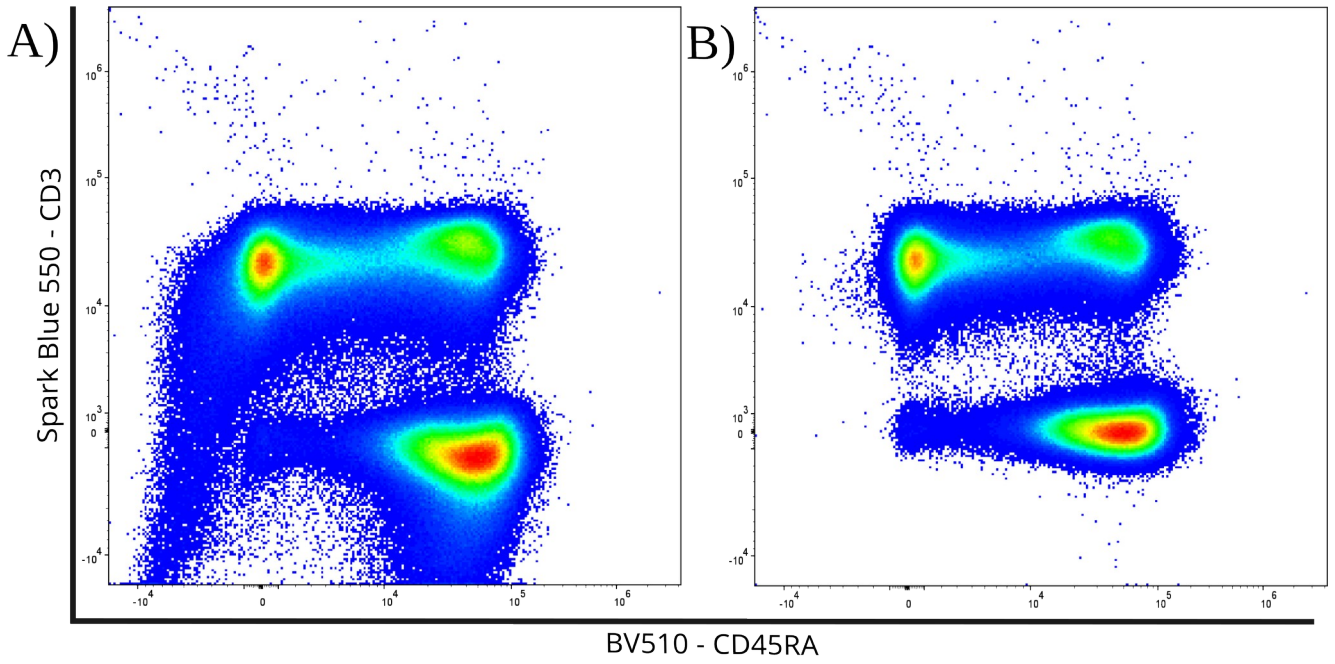



What fluorophores are present, and roughly how much



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# How to identify problematic unmixing controls




- Despite the critical importance unmixing controls play, few tools permit us to determine quality of the individual control before unmixing

- Existing quality control algorithms (PeacoQC, FlowAI, etc.) primarily focus on detecting laser and fluidic issues in already acquired .fcs files.

- Unmixing errors can be introduced by tandem degradational, additional autofluorescences, undiagnosed instrumental issues, etc.

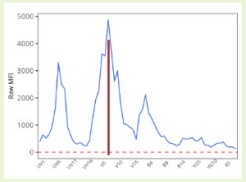


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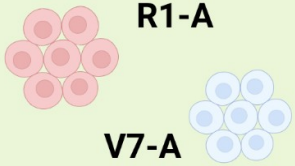
# Luciernaga




Code



Identify detector with brightest MFI for each cell




Group cells with shared peak detectors



Subtract Out Autofluorescence

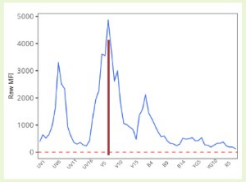
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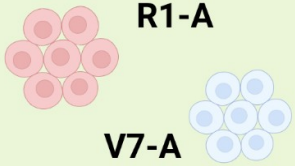
# Luciernaga




Code



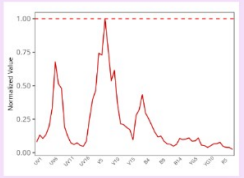
Identify detector with brightest MFI for each cell



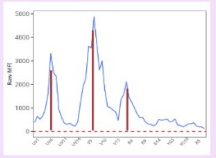

Group cells with shared peak detectors



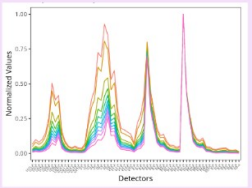
Subtract Out Autofluorescence



Normalize individual cell fluorescent signatures



Identify peaks using local maxima




Group cells with shared peaks & heights

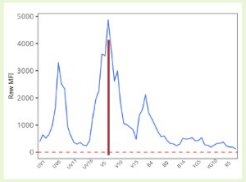


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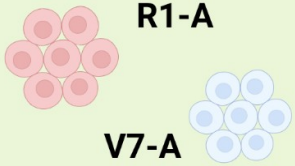
# Luciernaga




Code



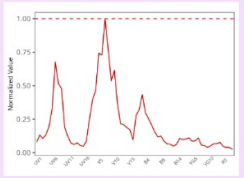
Identify detector with brightest MFI for each cell



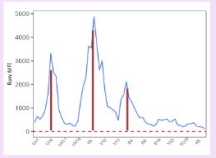

Group cells with shared peak detectors



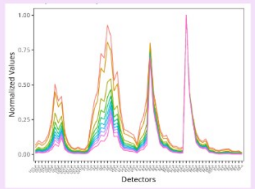
Subtract Out Autofluorescence



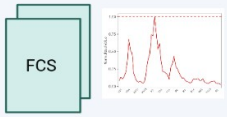
Normalize individual cell fluorescent signatures



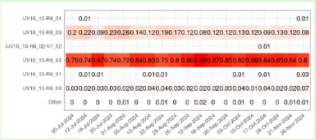
Identify peaks using local maxima



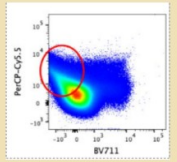
Group cells with shared peaks & heights




Enumerate and export out cells that share signature as .fcs files



Checks for tandem degradation

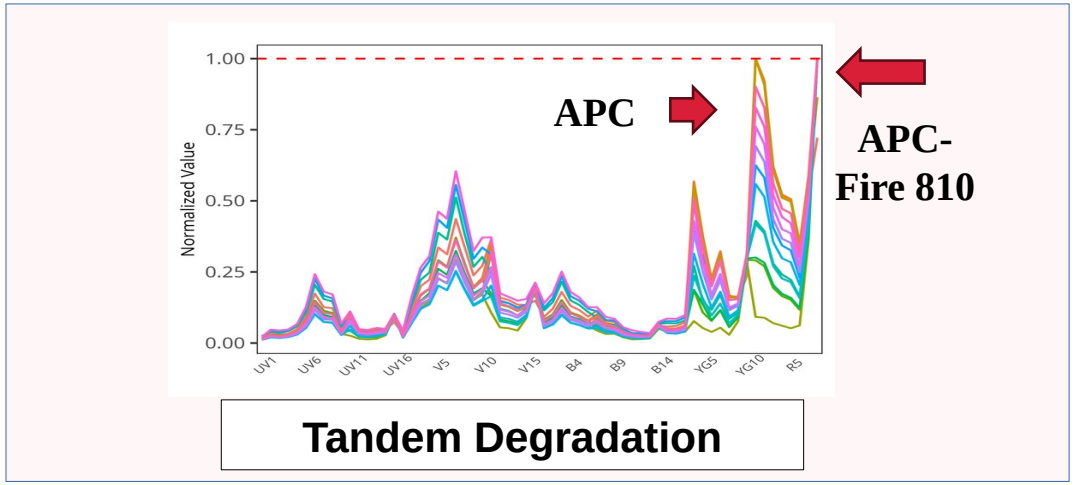


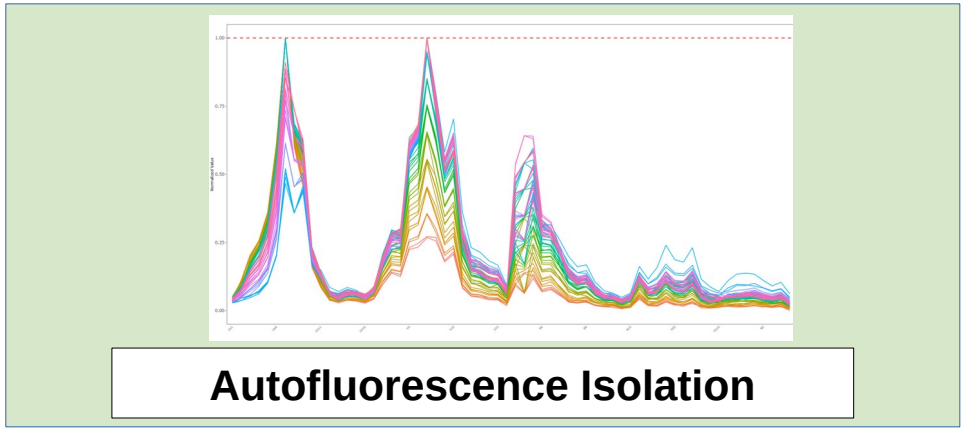
Autofluorescence Isolation

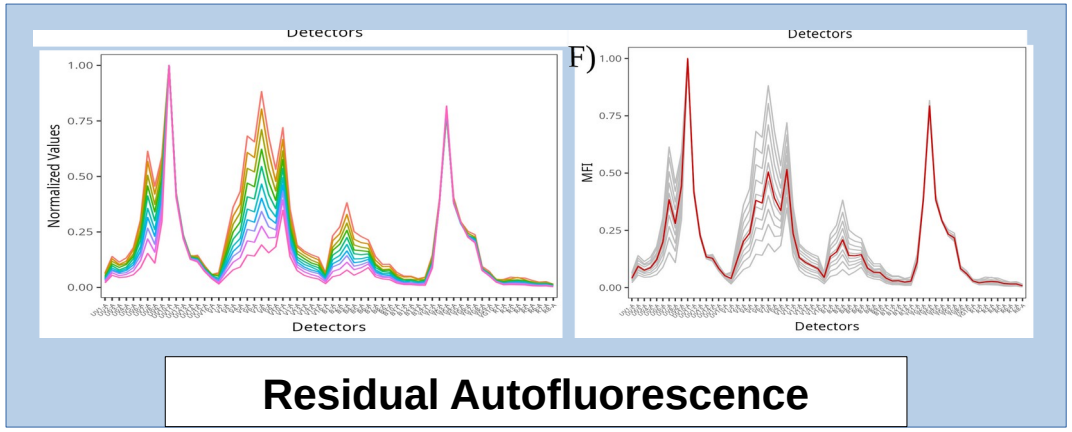
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# Automated tools for screening unmixing controls





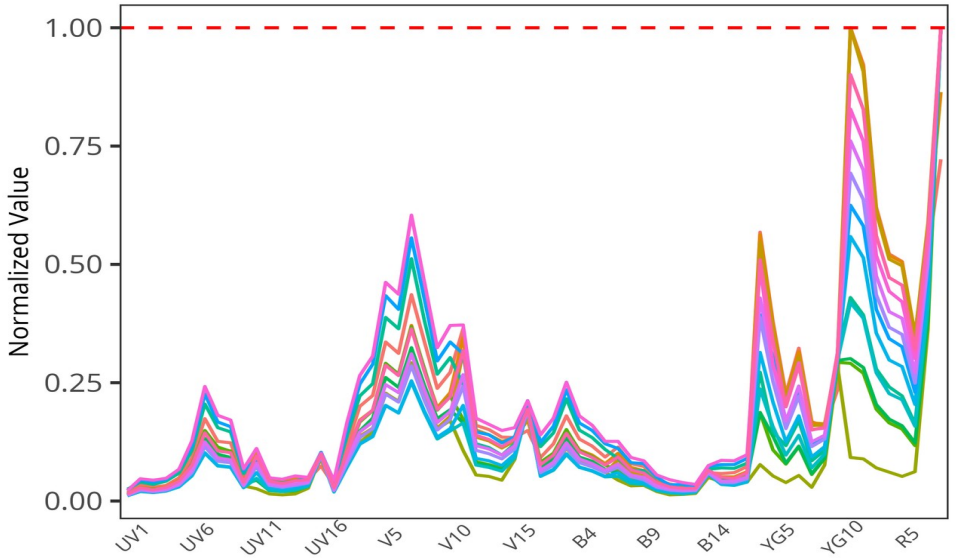

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# Identifying degrading fluorophores

Fluorophores in good shape → **one predominant normalized signature** within the unmixing control raw .fcs file

- When the fluorophore degrades, noticeable **increase in variant signatures**, making up substantial proportion of all cells in the unmixing control

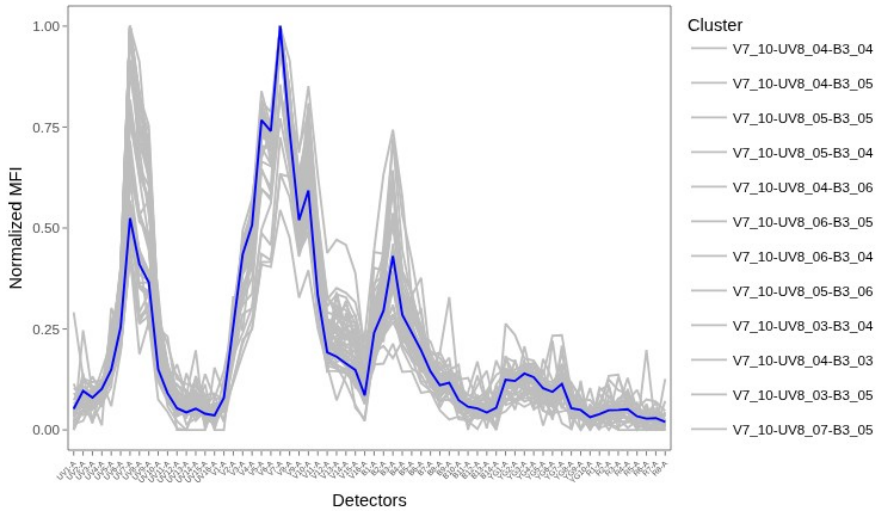
- Automated screening all acquired unmixing controls from campus users during off-hours

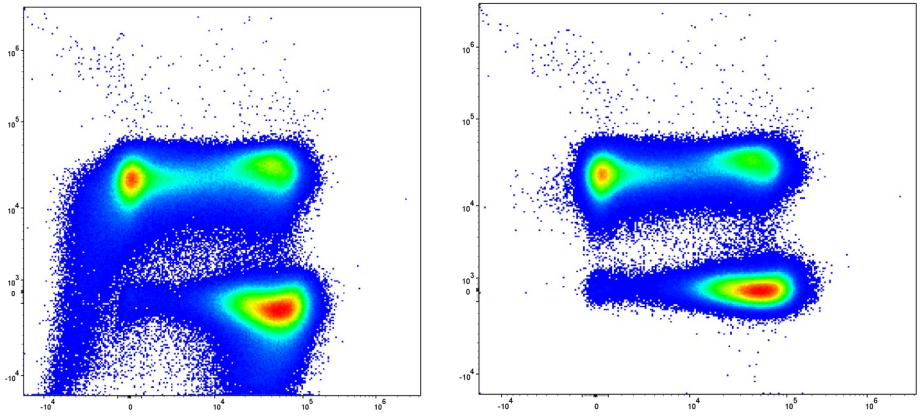





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# Isolating additional autofluorescence signatures



- This approach also works to isolate variant autofluorescence signatures from unstained controls

- Unlike other approaches, the isolated cells share normalized signatures, only differing in terms of brightness

- Return directly as an .fcs file containing just cells with same signature for use as an autofluorescent tag in SpectroFlo



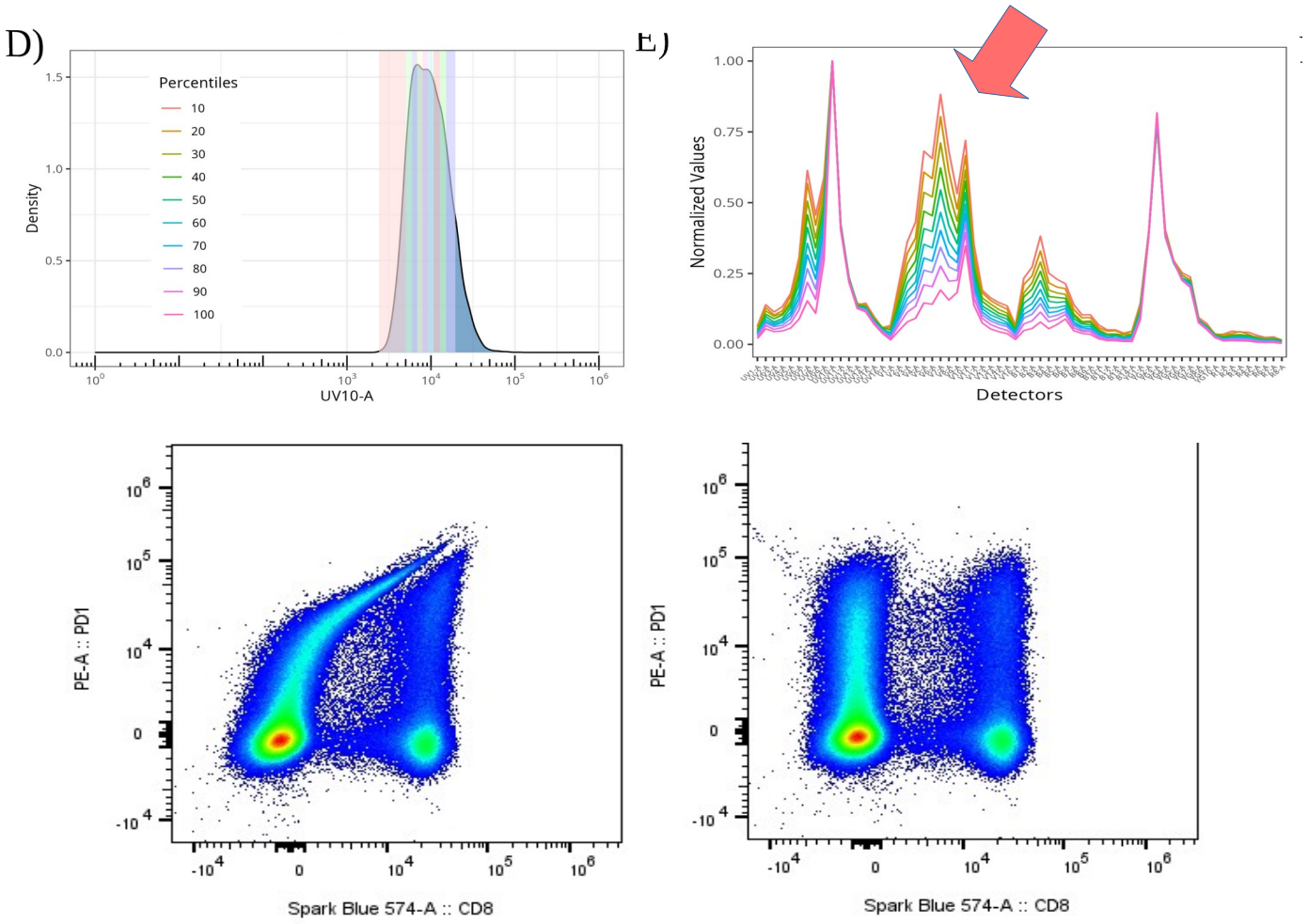
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# Residual autofluorescence in dim cell single-color controls

- **“As bright or brighter”**

- Dim antigen-fluorophore combinations acquired as **cell single-color controls** often have residual autofluorescence contribution to their signature

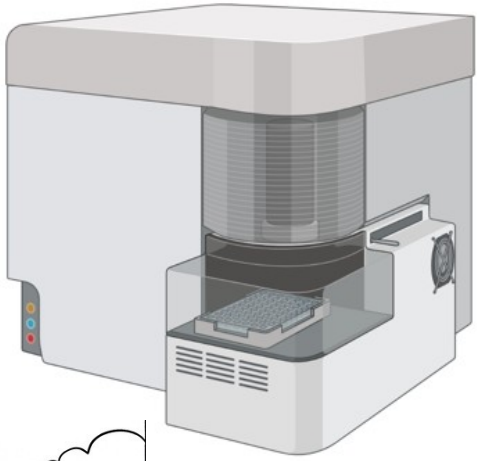
- Identifying and **switching these** dim cell single-color controls to **bead single-color controls** significantly improved unmixing for our panels

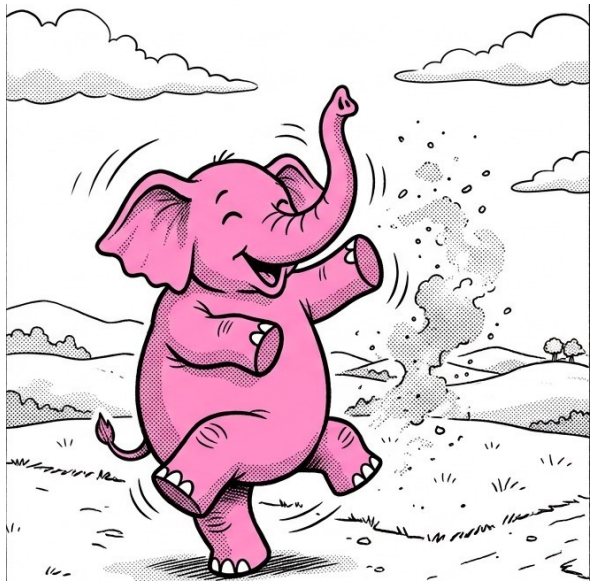



**Signature with residual AF**

**Signature without residual AF**

# To unlock the comprehensive biological discovery potential, we need to address these pink elephants





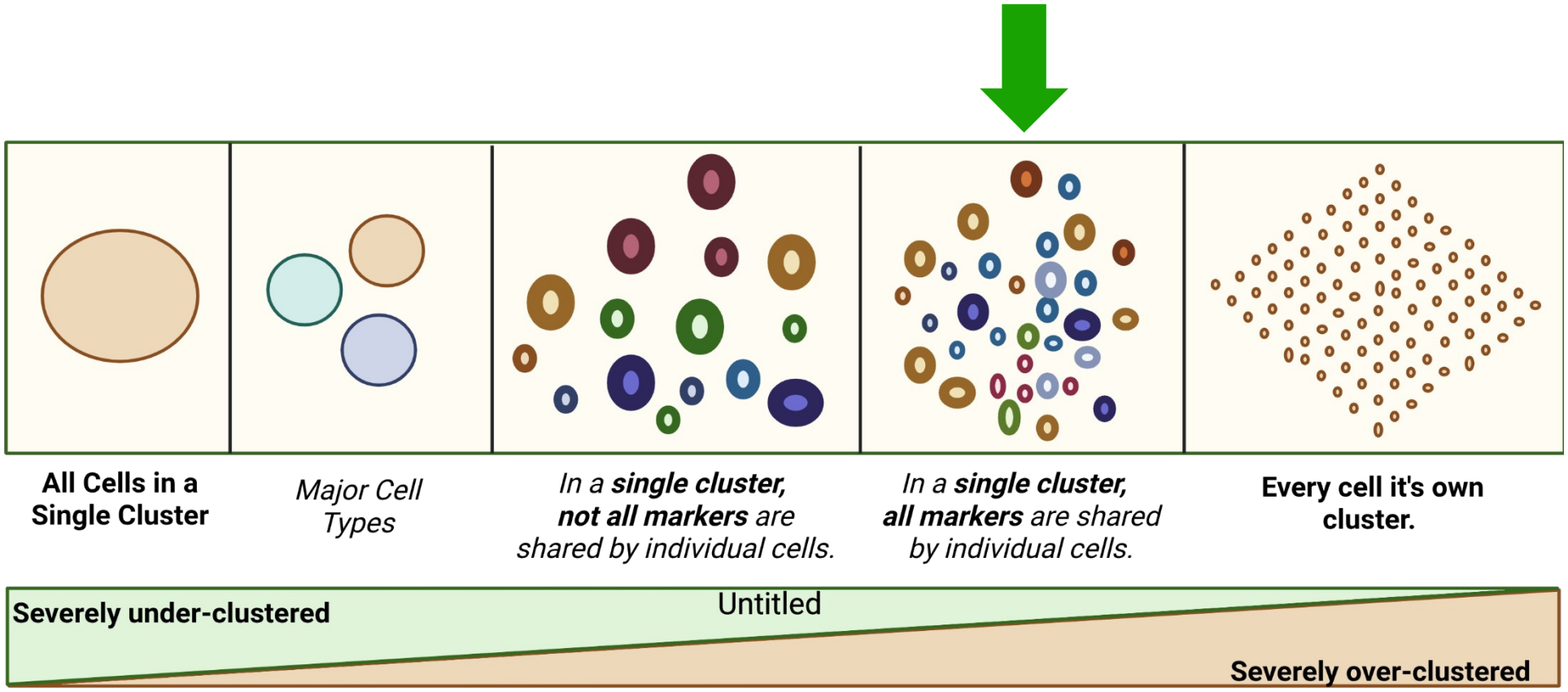
- [x] Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

- [ ] If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



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# Grouping cells that share every marker



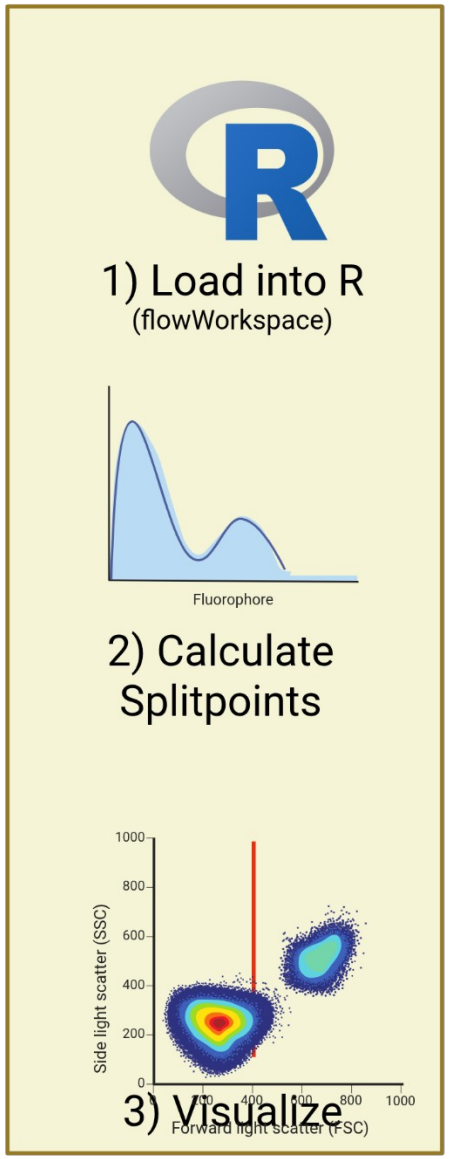

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# Coereba



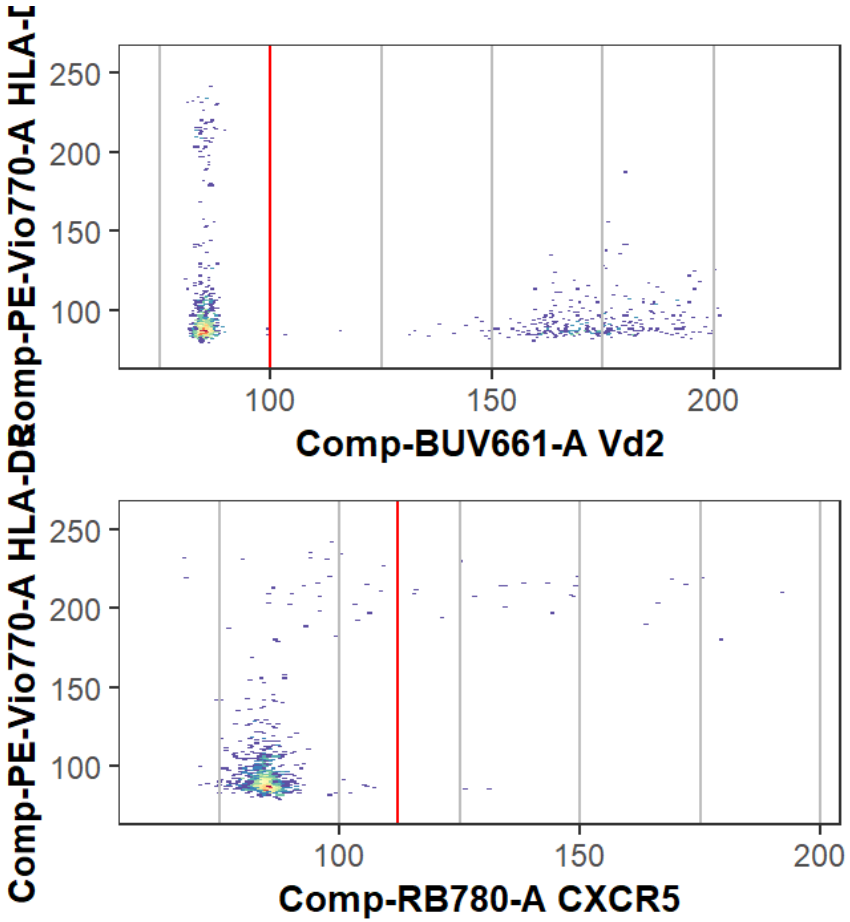

Code




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# Initial automated gate, with ability to adjust




- ❑ Initial gates for designated markers calculated via openCyto (mindensity)
- ❑ Ability to update gate_range constraints
- ❑ Launch a Shiny app and designate interactively

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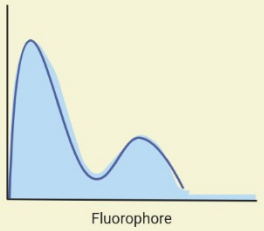
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# Coereba

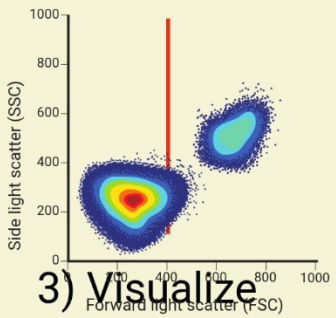



Code

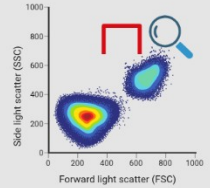
1) Load into R
(flowWorkspace)




2) Calculate Splitpoints

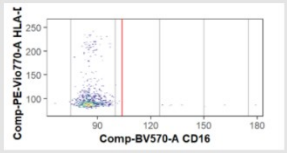



3) Visualize




4) Gate Constraints

5) Shiny App adjustments



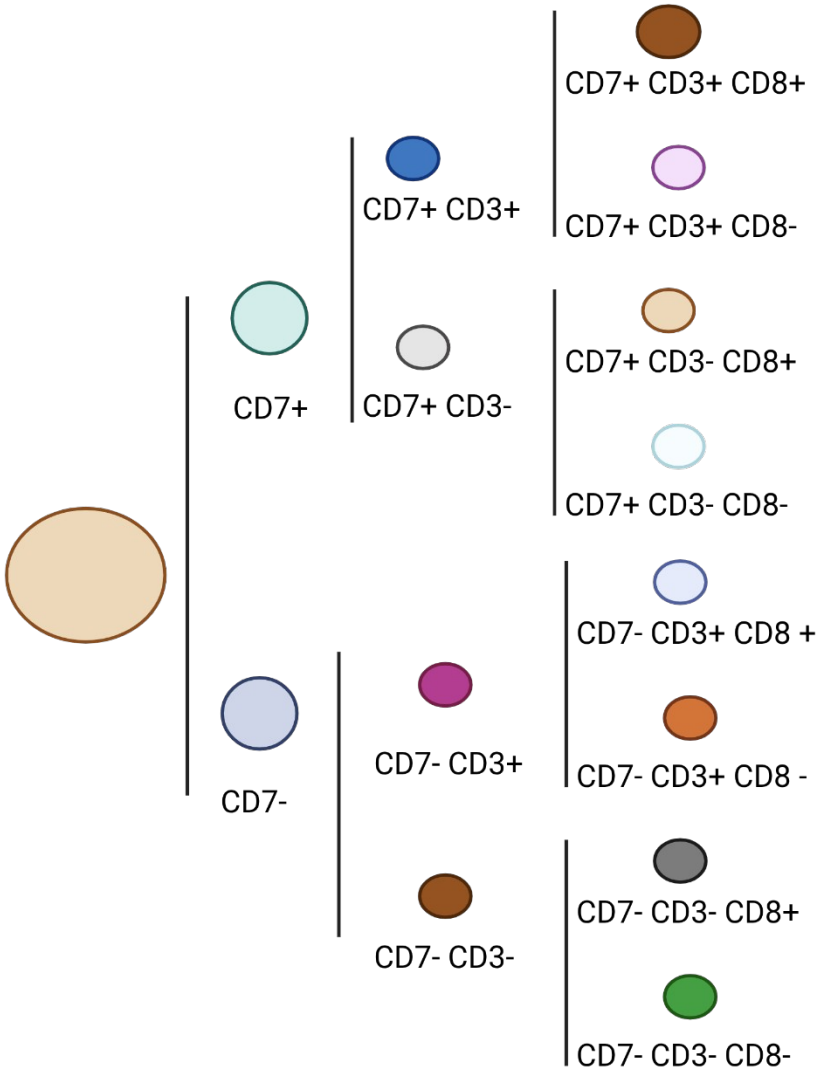

6) Splitpoints updated

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# CoerebaID column gets added to the .fcs file

| | | | | | |
|---|---|---|---|---|---|
| PEVio770pos | APCneg | AlexaFluor647neg | APCR700neg | APCFire750neg | APCFire810neg |
| PEVio770pos | APCneg | AlexaFluor647neg | APCR700neg | APCFire750neg | APCFire810neg |
| PEVio770pos | APCpos | AlexaFluor647neg | APCR700neg | APCFire750pos | APCFire810neg |
| PEVio770pos | APCneg | AlexaFluor647pos | APCR700neg | APCFire750neg | APCFire810pos |



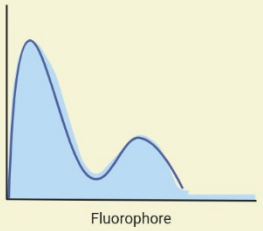

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# Coereba

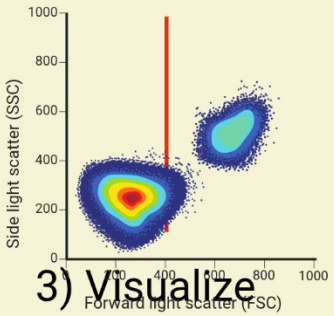



Code

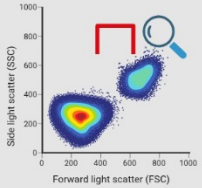
1) Load into R
(flowWorkspace)




2) Calculate Splitpoints

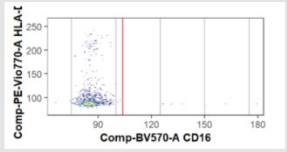



3) Visualize

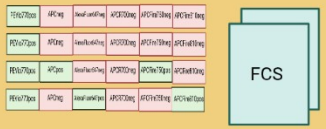



4) Gate Constraints

5) Shiny App adjustments

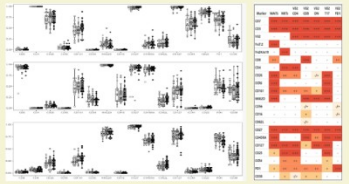



6) Splitpoints updated




7) CoerebaID

8) Summarized Experiment

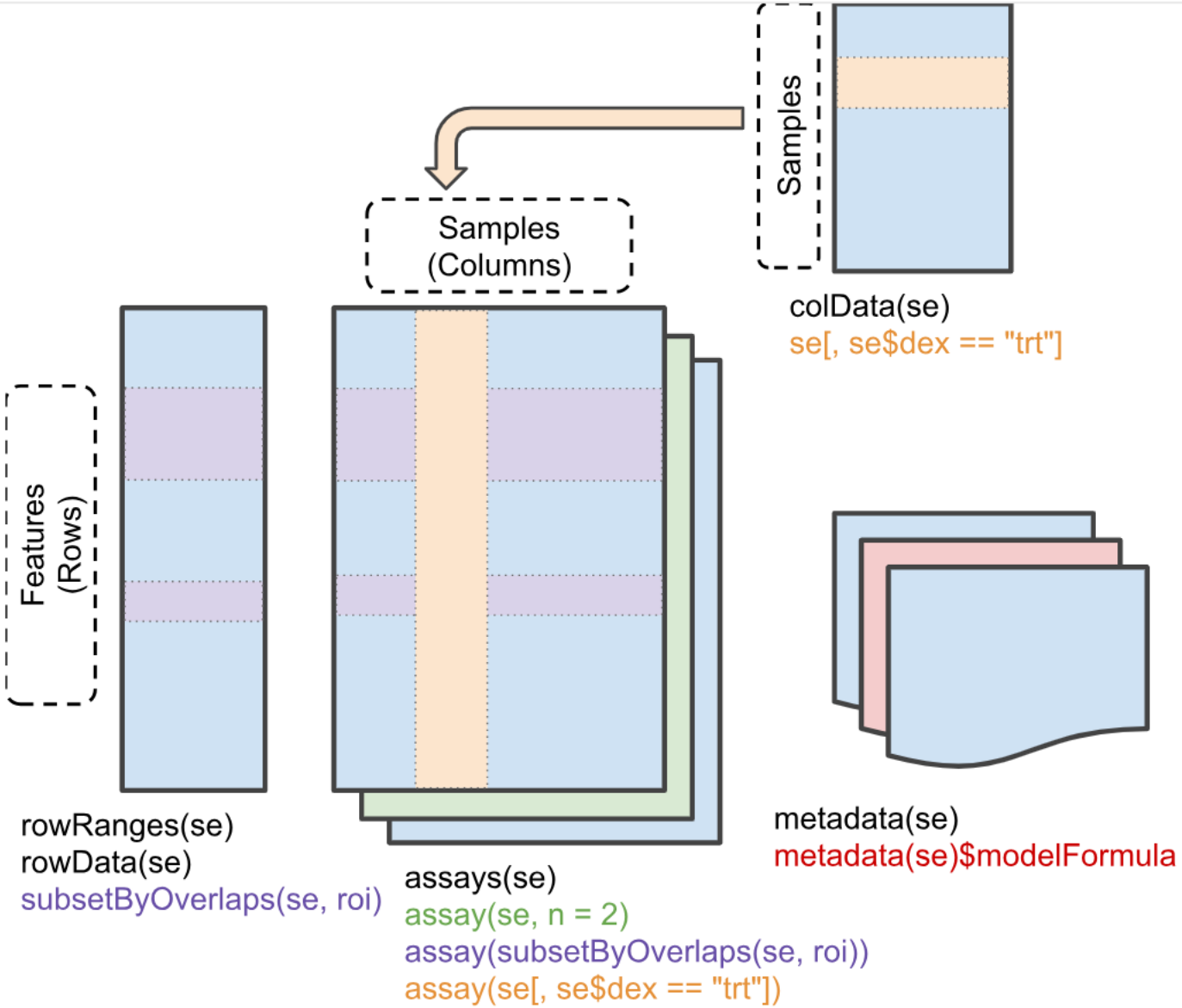


9) Visualize

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# Summarized Experiment

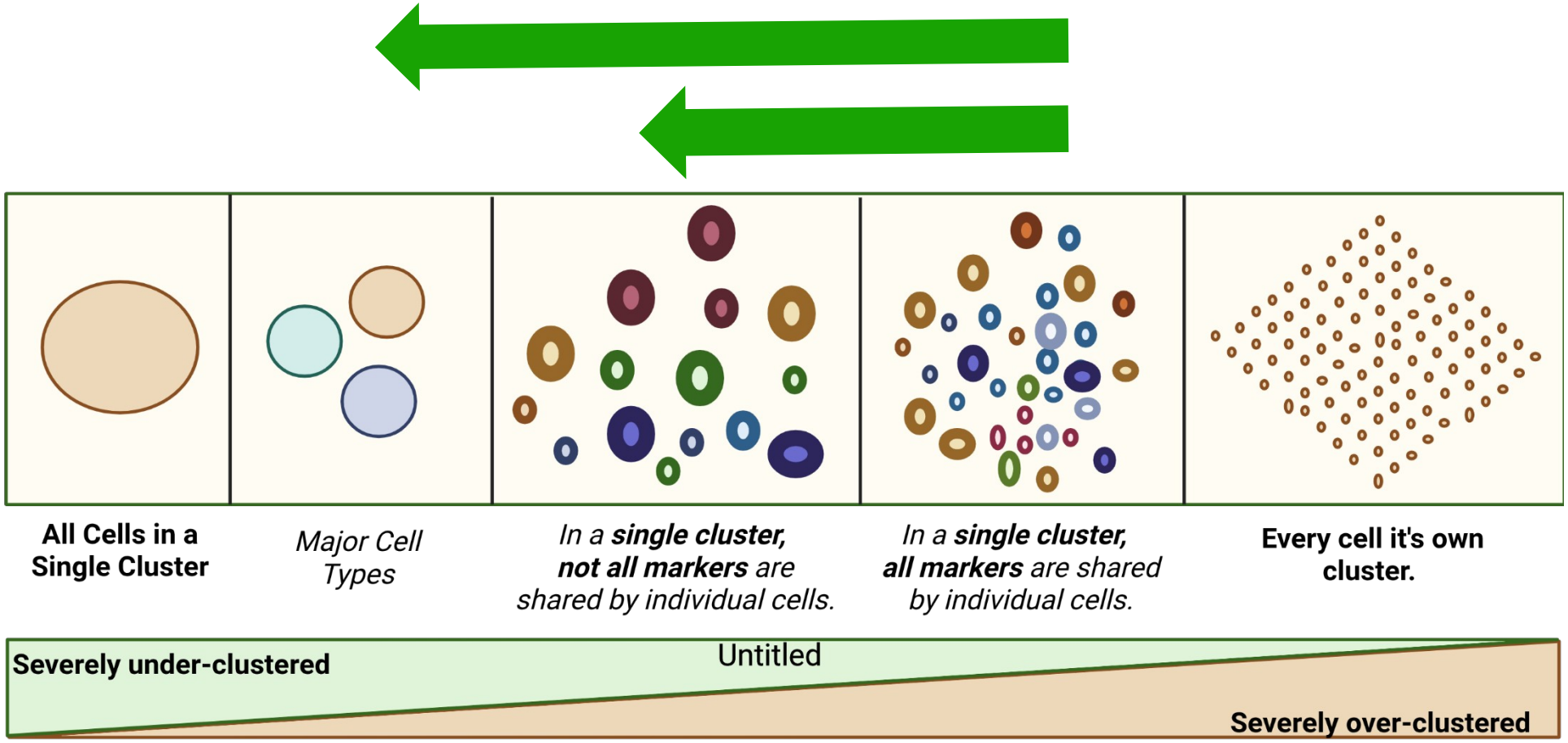



- Permits rapid re-aggregation of the terminal nodes for re-evaluation



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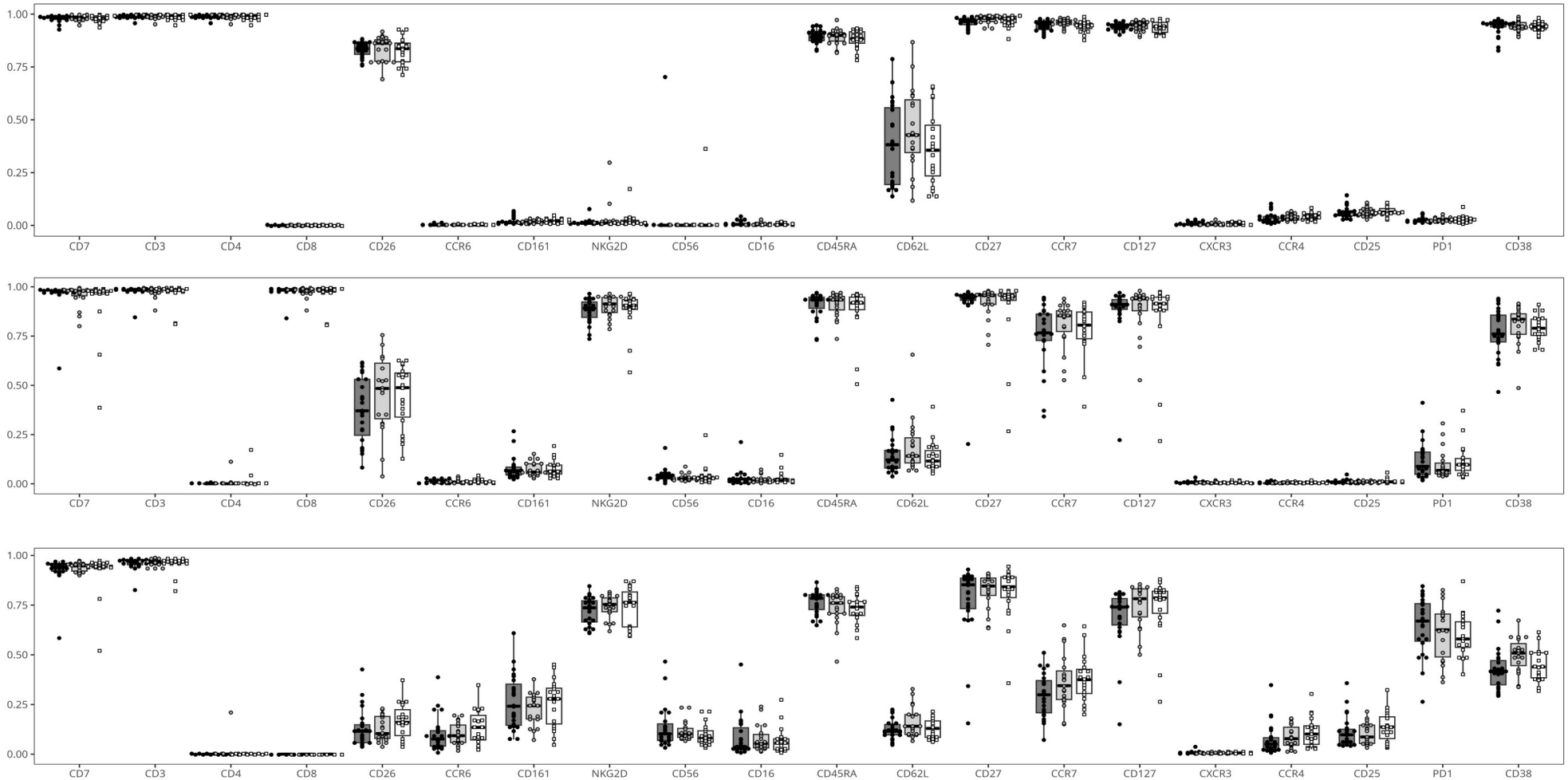
# Selectively re-aggregate in seconds






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# Retrieve marker expression by individual donor

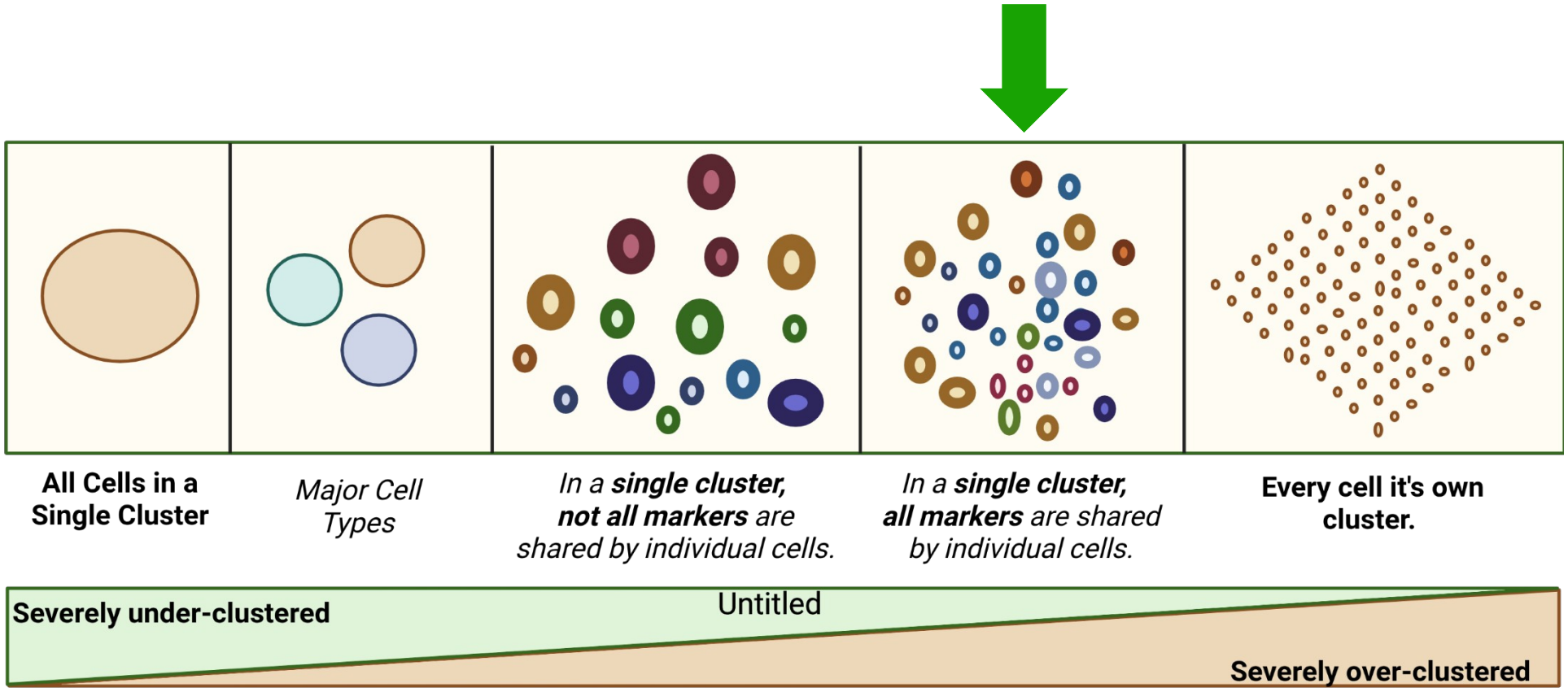



- ❑ **CD4 T cells**
- ❑ **CD8 T cells**
- **DN T cells**

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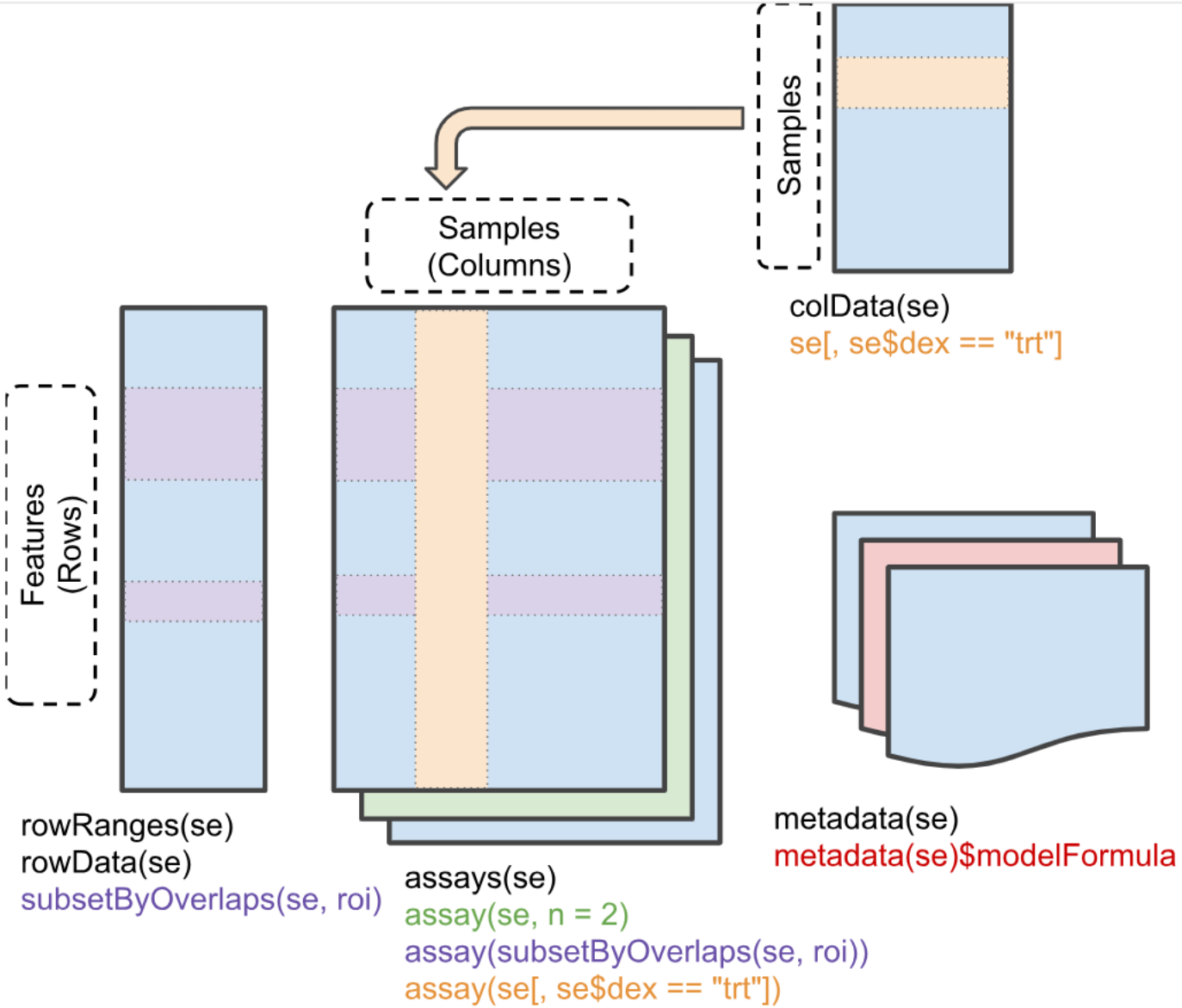
# Grouping cells that share every marker




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# Summarized Experiment




- ❑ Permits rapid re-aggregation of the terminal nodes for re-evaluation
- ❑ Portable. Don't like how we aggregated groups of cells? File is small enough to store in the same data repository as .fcs files for subsequent re-analysis.
- ❑ Filterable by metadata or node.
  - Nodes only in HEU-hi?
  - Nodes unique to one individual?
- ❑ Ties in with downstream Bioconductor statistics packages



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# Biology is (fortunately) not just math




- **29-color** SFC panel
  → **536 million (theory)**

**When actually exploring our panels dimensional space**

**CD4 T cells → 16241**

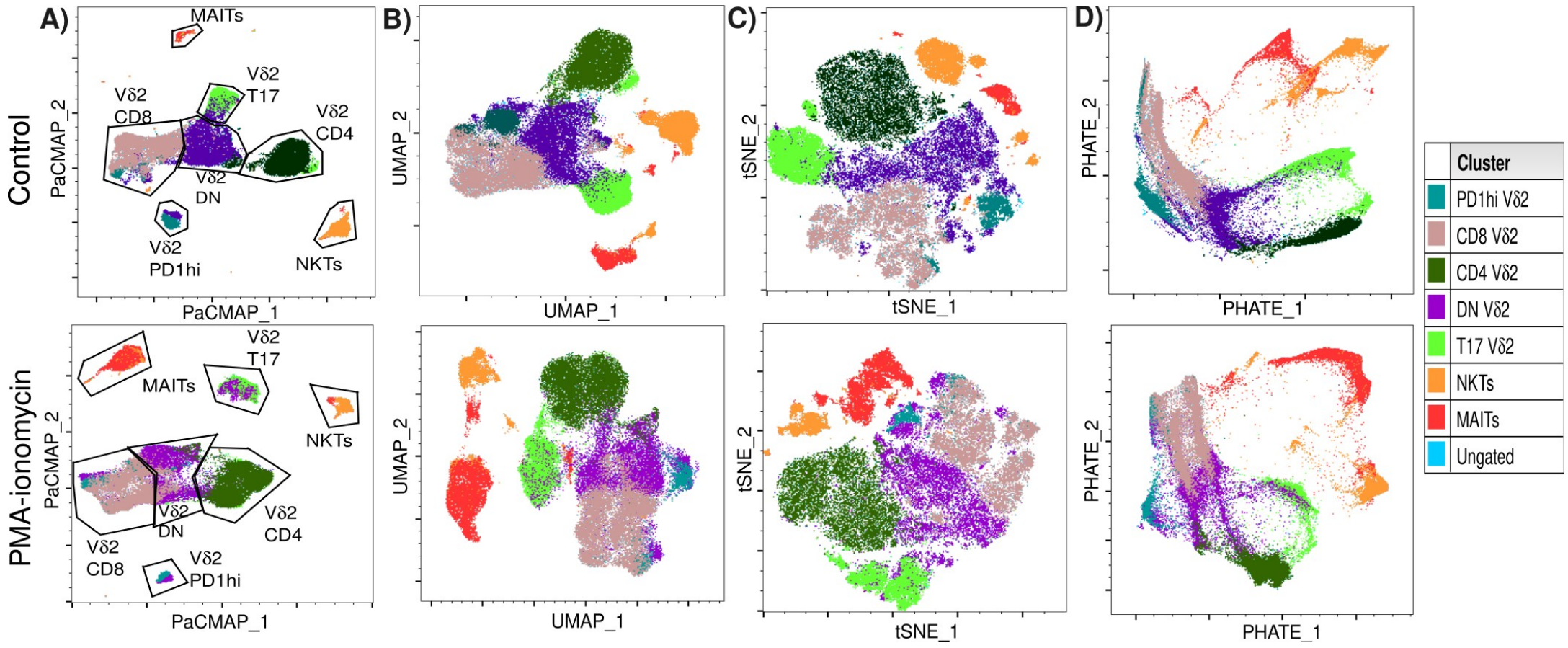
**CD8 T cells → 12466**

**DN T cells → 9035**



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# Ability to contrast unsupervised methods vs. “manual” ground-truth




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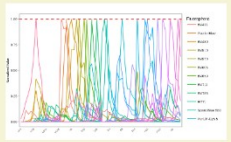
# Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



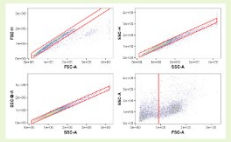

2) Unmixing Controls
(Luciernaga)



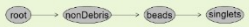
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



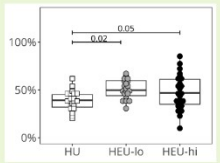
5) Clean Up Gates
(FlowJo or flowWorkspace)




6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)




8) Major subset Outputs

Iterating by Experiment



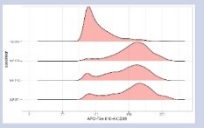

9) Subset Export



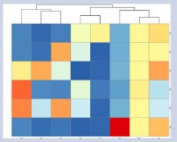
10) Return to R
(flowWorkspace)



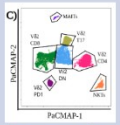

11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)




Summarized Experiment



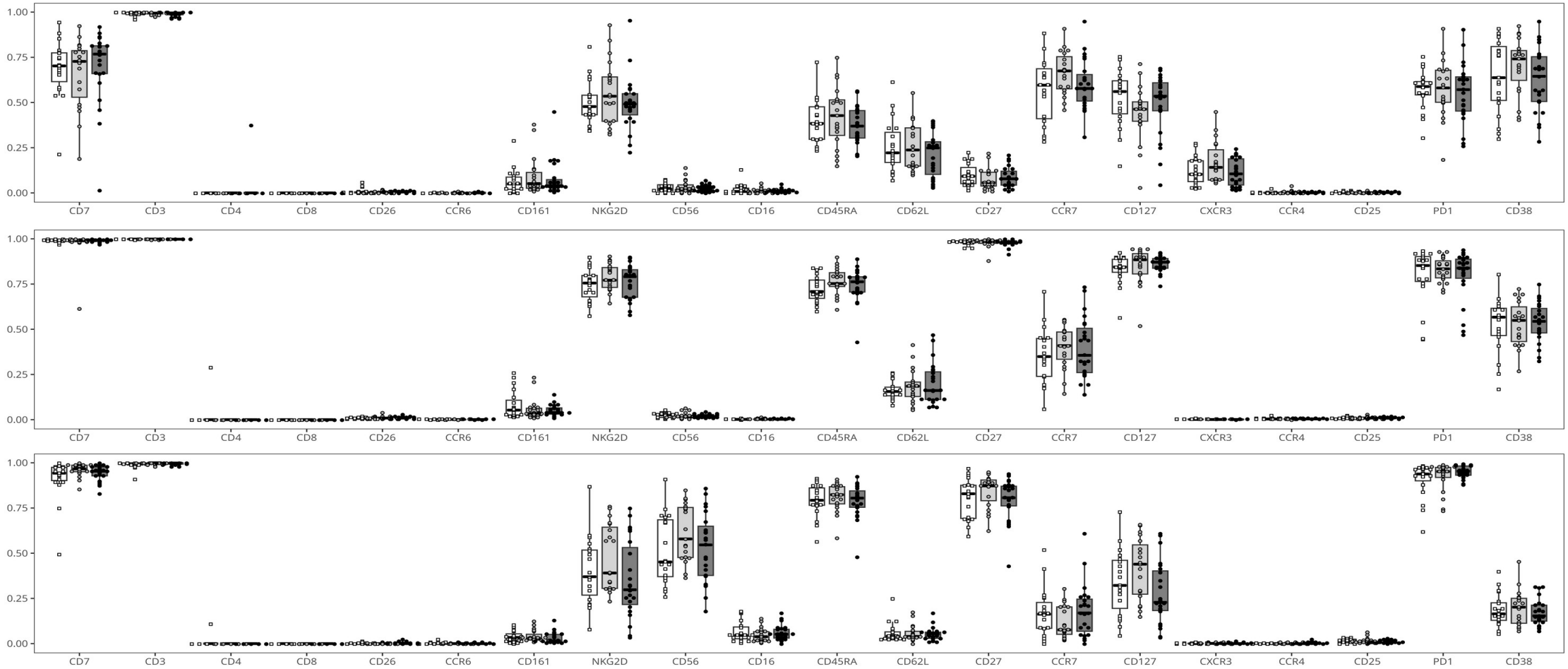

15) Coereba

Iterating by Cell Subset



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# For each island, re-aggregate cells from each donor




**DN T cells**

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# Cord blood T cells - Baseline

| Marker | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| CD7 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD3 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| VD2 | - | - | - | - | - | - | - | - |
| Va7.2 | - | - | - | - | - | - | - | +++ |
| Va24Ja18 | - | - | - | - | - | - | - | - |
| CD8 | - | - | - | - | - | - | - | - |
| CD4 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD26 | -/+ | ++ | + | ++ | +++ | +++ | +++ | +++ |
| CCR6 | - | - | - | - | - | - | - | - |
| CD161 | - | + | - | - | - | - | ++ | - |
| NKG2D | - | - | -/+ | - | - | - | +++ | - |
| CD56 | - | - | - | - | - | - | +++ | - |
| CD16 | - | - | - | - | - | - | + | - |
| CD62L | -/+ | - | - | +++ | - | +++ | + | + |
| CD27 | +++ | +++ | + | +++ | +++ | +++ | +++ | +++ |
| CD45RA | +++ | + | +++ | +++ | +++ | +++ | +++ | +++ |
| CD127 | - | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD25 | +++ | +++ | - | - | - | - | - | - |
| CCR4 | - | +++ | - | - | - | - | - | - |
| PD1 | -/+ | + | - | - | - | - | - | - |
| CD38 | +++ | + | +++ | +++ | +++ | +++ | +++ | +++ |

**CD4 T cells**

| Marker | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CD7 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | ++ | +++ |
| CD3 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| VD2 | - | - | - | - | - | - | - | - | - | - | - |
| Va7.2 | +++ | - | - | - | - | - | - | - | - | - | - |
| Va24Ja18 | - | - | - | - | - | - | - | - | - | - | - |
| CD8 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD4 | - | - | - | - | - | - | - | - | - | - | - |
| CD26 | ++ | - | + | ++ | -/+ | +++ | +++ | - | - | - | - |
| CCR6 | - | - | - | - | - | - | +++ | - | - | - | - |
| CD161 | - | ++ | - | - | - | +++ | +++ | - | - | -/+ | +++ |
| NKG2D | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ | - | +++ | +++ |
| CD56 | - | - | - | - | - | - | - | - | -/+ | - | +++ |
| CD16 | - | +++ | - | - | - | - | - | - | - | - | +++ |
| CD62L | - | - | +++ | - | ++ | - | - | -/+ | - | - | - |
| CD27 | +++ | + | +++ | +++ | + | +++ | +++ | +++ | +++ | + | -/+ |
| CD45RA | +++ | +++ | +++ | +++ | +++ | +++ | +++ | ++ | ++ | +++ | +++ |
| CD127 | +++ | - | +++ | +++ | +++ | +++ | +++ | +++ | -/+ | -/+ | - |
| CD25 | - | - | - | - | - | - | + | - | - | - | - |
| CCR4 | - | - | - | - | - | - | + | - | - | - | - |
| PD1 | - | ++ | - | - | - | - | - | + | +++ | + | - |
| CD38 | +++ | +++ | +++ | +++ | +++ | +++ | - | -/+ | + | ++ | +++ |

**CD8 T cells**

| Marker | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| CD7 | ++ | +++ | +++ | +++ | +++ |
| CD3 | +++ | +++ | +++ | +++ | +++ |
| VD2 | - | - | - | - | - |
| Va7.2 | - | - | - | - | - |
| Va24Ja18 | - | - | - | - | - |
| CD8 | - | - | - | - | - |
| CD4 | - | - | - | - | - |
| CD26 | - | - | - | +++ | - |
| CCR6 | - | - | - | +++ | - |
| CD161 | - | - | - | +++ | +++ |
| NKG2D | ++ | +++ | + | +++ | +++ |
| CD56 | - | - | ++ | - | +++ |
| CD16 | - | - | - | - | +++ |
| CD62L | -/+ | -/+ | - | - | - |
| CD27 | - | +++ | +++ | +++ | -/+ |
| CD45RA | + | +++ | +++ | +++ | +++ |
| CD127 | ++ | +++ | + | +++ | - |
| CD25 | - | - | - | ++ | - |
| CCR4 | - | - | - | ++ | - |
| PD1 | ++ | +++ | +++ | -/+ | - |
| CD38 | ++ | ++ | -/+ | - | +++ |

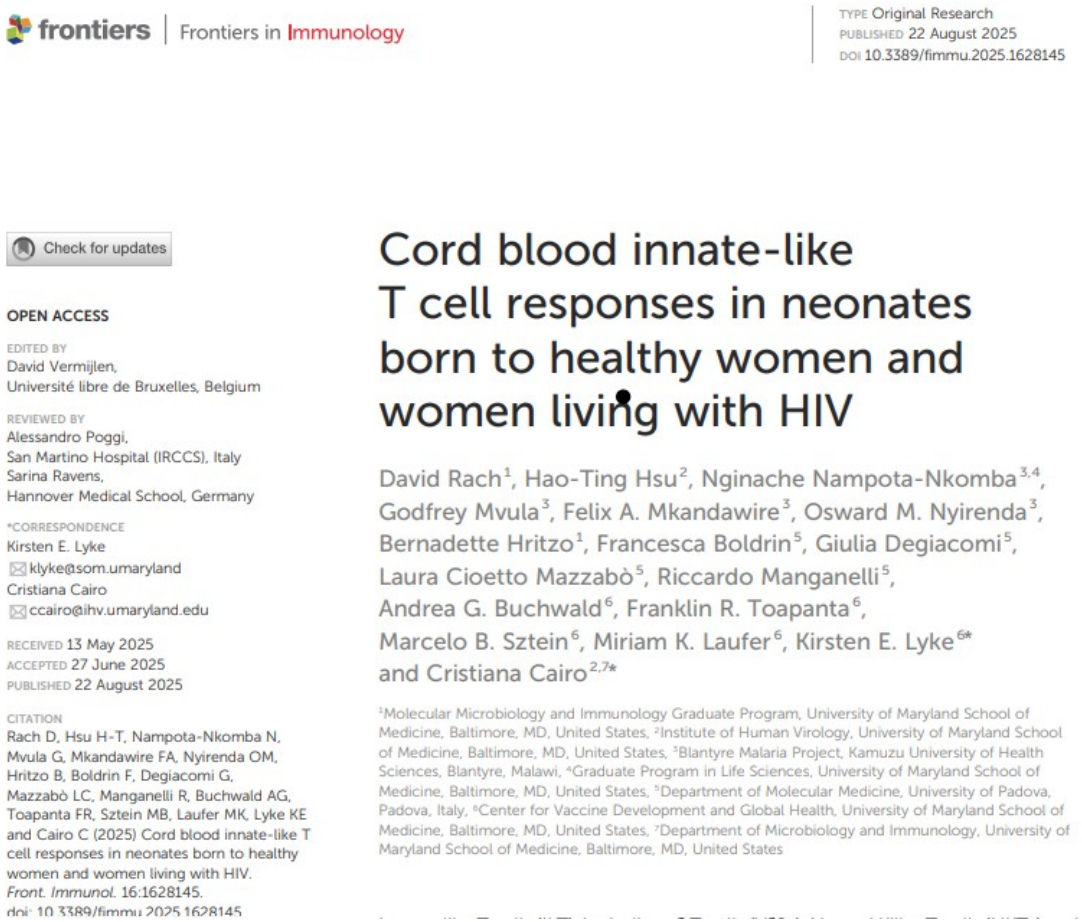
**DN T cells**

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# Published in Frontiers in Immunology

Paper



frontiers | Frontiers in Immunology

TYPE Original Research
PUBLISHED 22 August 2025
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Check for updates

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## Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV

David Rach[1], Hao-Ting Hsu[2], Nginache Nampota-Nkomba[3,4], Godfrey Mvula[3], Felix A. Mkandawire[3], Osward M. Nyirenda[3], Bernadette Hritzo[1], Francesca Boldrin[5], Giulia Degiacomi[5], Laura Cioetto Mazzabò[5], Riccardo Manganelli[5], Andrea G. Buchwald[6], Franklin R. Toapanta[6], Marcelo B. Sztein[6], Miriam K. Laufer[6], Kirsten E. Lyke[6*] and Cristiana Cairo[2,7*]

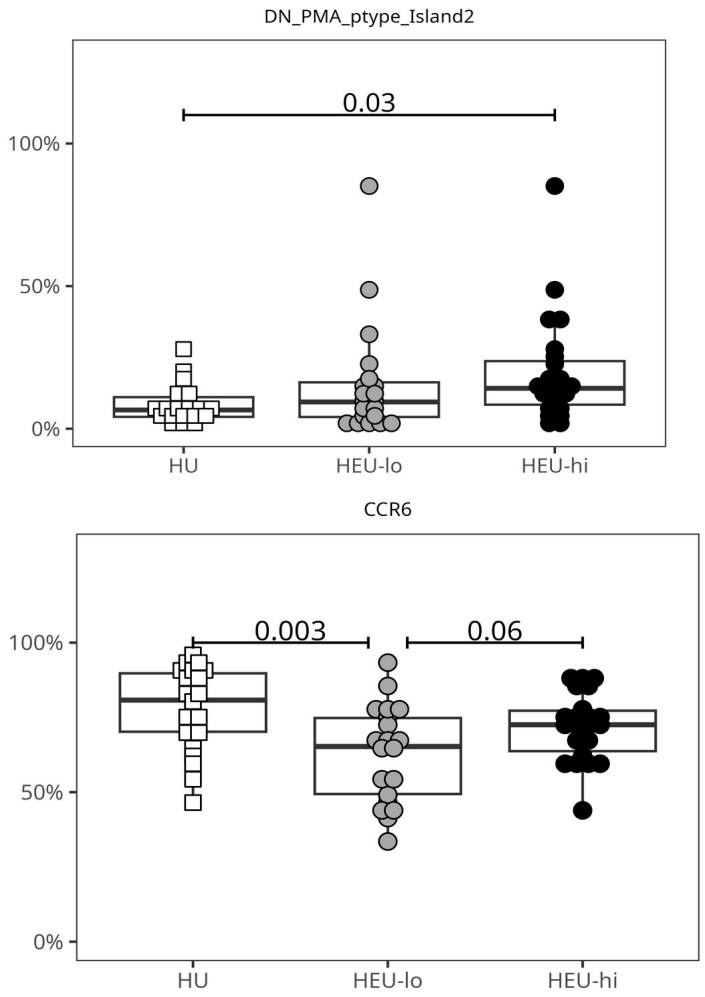
[1]Molecular Microbiology and Immunology Graduate Program, University of Maryland School of Medicine, Baltimore, MD, United States, [2]Institute of Human Virology, University of Maryland School of Medicine, Baltimore, MD, United States, [3]Blantyre Malaria Project, Kamuzu University of Health Sciences, Blantyre, Malawi, [4]Graduate Program in Life Sciences, University of Maryland School of Medicine, Baltimore, MD, United States, [5]Department of Molecular Medicine, University of Padova, Padova, Italy, [6]Center for Vaccine Development and Global Health, University of Maryland School of Medicine, Baltimore, MD, United States, [7]Department of Microbiology and Immunology, University of Maryland School of Medicine, Baltimore, MD, United States

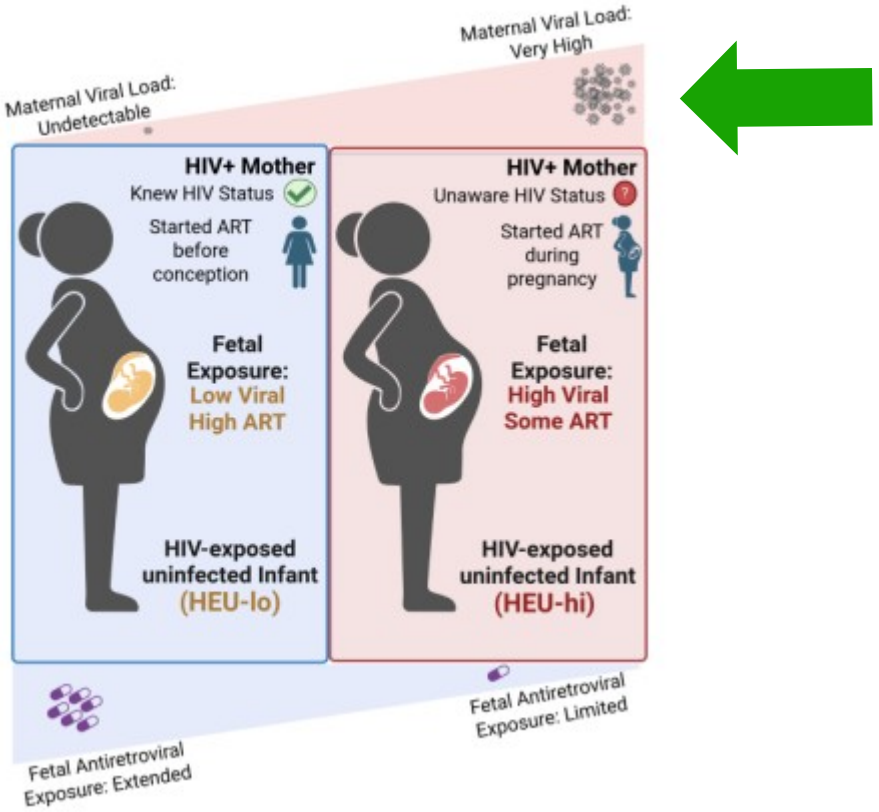


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# CD4-CD8- Cord blood T cell subsets driving observed IFNɣ increases

| Marker | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| CD7 | +++ | +++ | +++ | +++ |
| CD3 | +++ | +++ | +++ | +++ |
| VD2 | - | - | - | - |
| Va7.2 | - | - | - | - |
| Va24Ja18 | - | - | - | - |
| CD8 | - | - | - | - |
| CD4 | - | - | - | - |
| CD26 | +++ | - | - | - |
| CCR6 | ++ | - | - | - |
| CD161 | +++ | +++ | - | - |
| NKG2D | +++ | +++ | +++ | - |
| CD56 | - | ++ | - | -/+ |
| CD16 | - | - | - | - |
| CD62L | - | - | - | - |
| CD27 | +++ | - | +++ | ++ |
| CD45RA | +++ | +++ | +++ | +++ |
| CD127 | +++ | - | +++ | - |
| CD25 | ++ | - | - | - |
| CCR4 | - | - | - | - |
| PD1 | ++ | -/+ | +++ | +++ |
| CD38 | - | +++ | ++ | + |
| CD69 | +++ | +++ | +++ | +++ |
| TNFα | + | ++ | - | - |
| IFNγ | - | ++ | - | -/+ |
| CD107a | - | - | - | - |



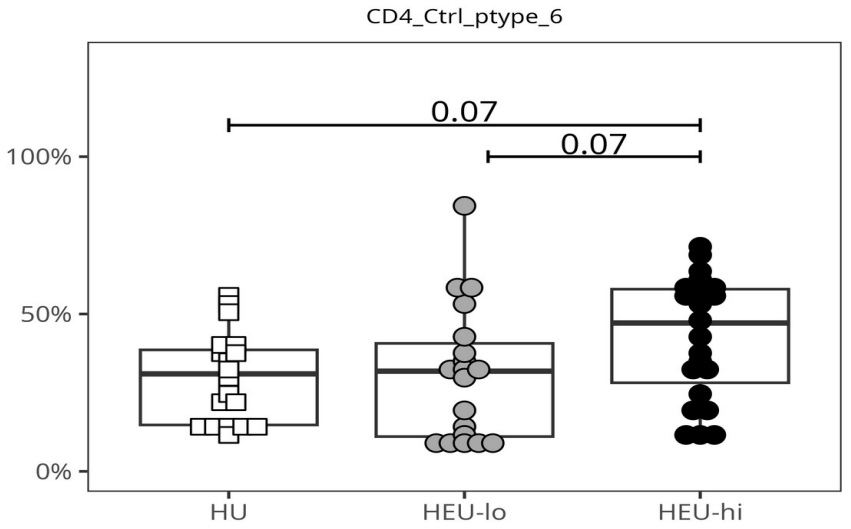



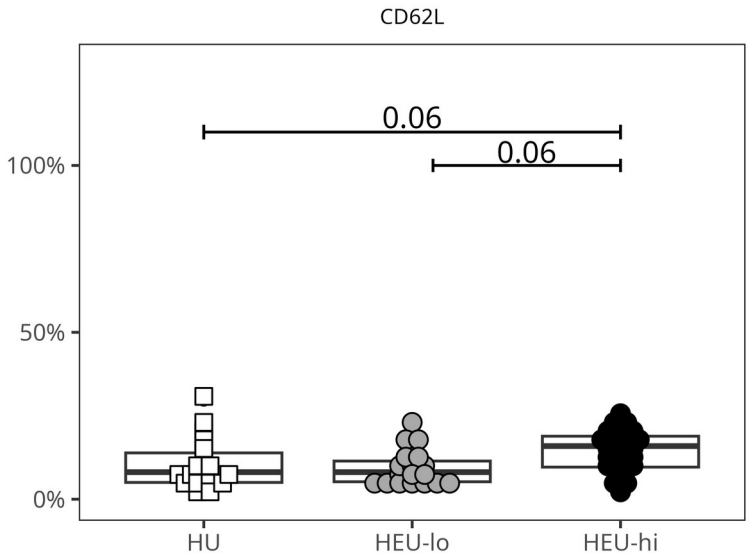

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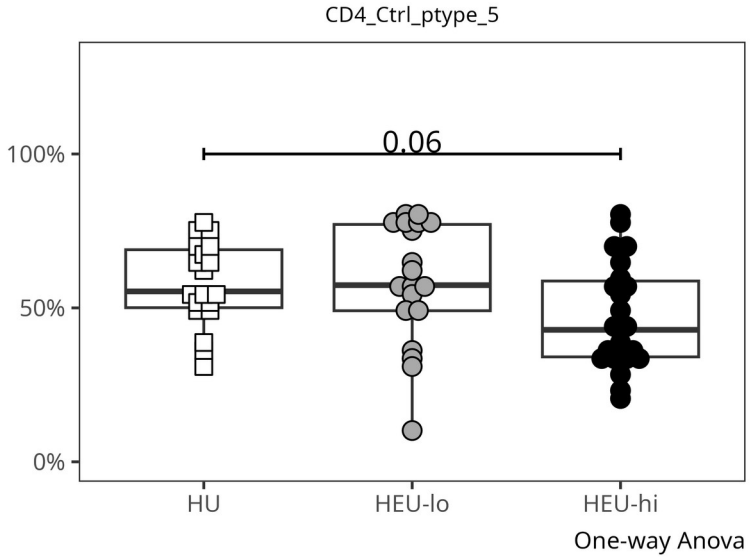
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# Conventional Cord blood T cell subsets also show changes in CD62L

| Marker | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| CD7 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD3 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| VD2 | - | - | - | - | - | - | - | - |
| Va7.2 | - | - | - | - | - | - | - | +++ |
| Va24Ja18 | - | - | - | - | - | - | - | - |
| CD8 | - | - | - | - | - | - | - | - |
| CD4 | +++ | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD26 | -/+ | ++ | + | ++ | +++ | +++ | +++ | +++ |
| CCR6 | - | - | - | - | - | - | - | - |
| CD161 | - | + | - | - | - | - | ++ | - |
| NKG2D | - | - | -/+ | - | - | - | +++ | - |
| CD56 | - | - | - | - | - | - | +++ | - |
| CD16 | - | - | - | - | - | - | + | - |
| CD62L | -/+ | - | - | +++ | - | +++ | + | + |
| CD27 | +++ | +++ | + | +++ | +++ | +++ | +++ | +++ |
| CD45RA | +++ | + | +++ | +++ | +++ | +++ | +++ | +++ |
| CD127 | - | +++ | +++ | +++ | +++ | +++ | +++ | +++ |
| CD25 | +++ | +++ | - | - | - | - | - | - |
| CCR4 | - | +++ | - | - | - | - | - | - |
| PD1 | -/+ | + | - | - | - | - | - | - |
| CD38 | +++ | + | +++ | +++ | +++ | +++ | +++ | +++ |



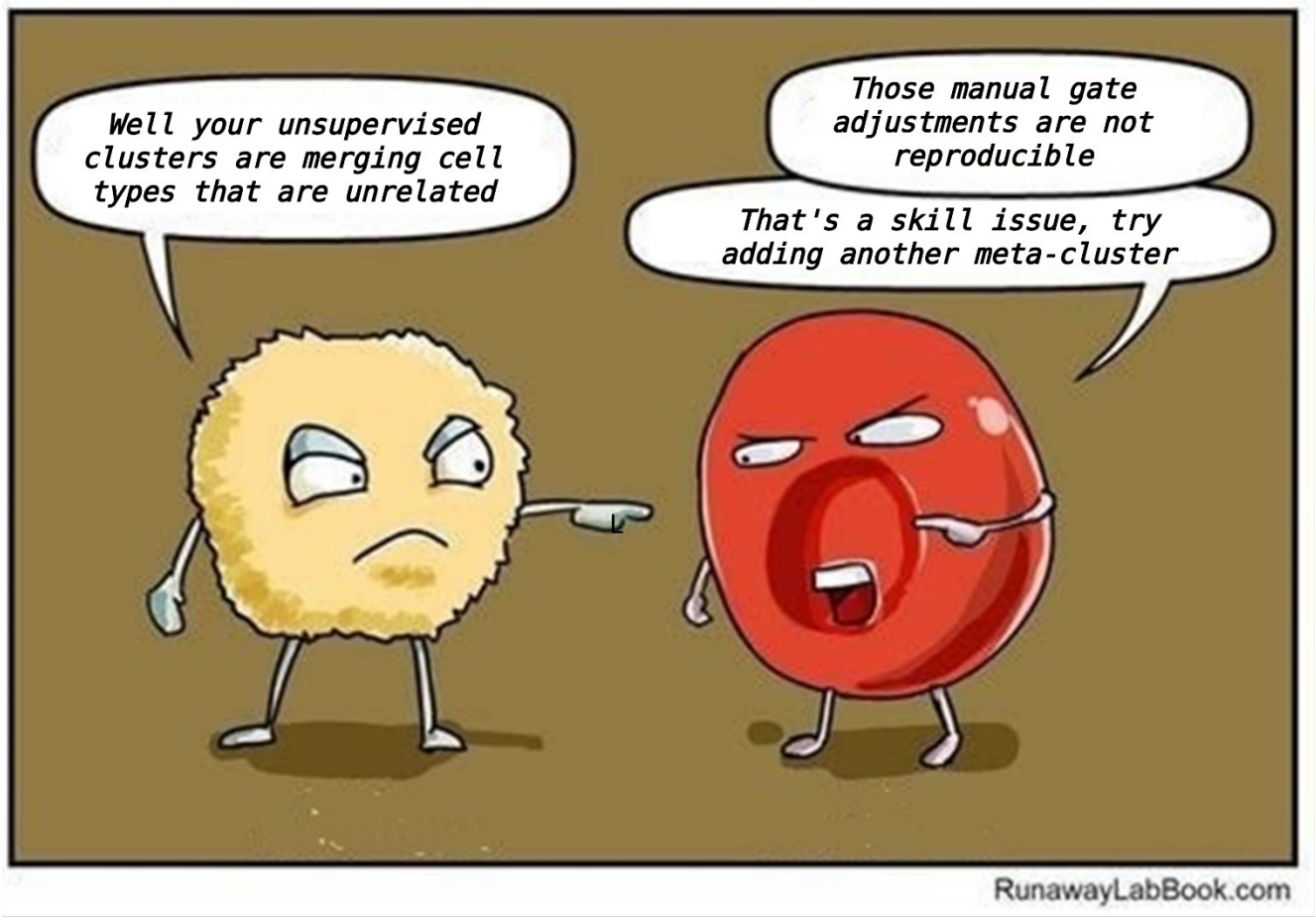






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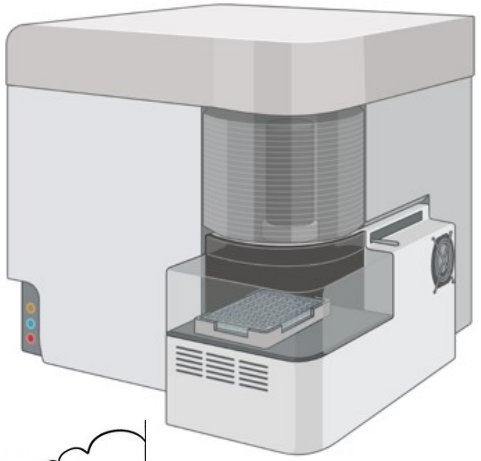
# Functional and Iterative: plenty for everyone to dislike :)

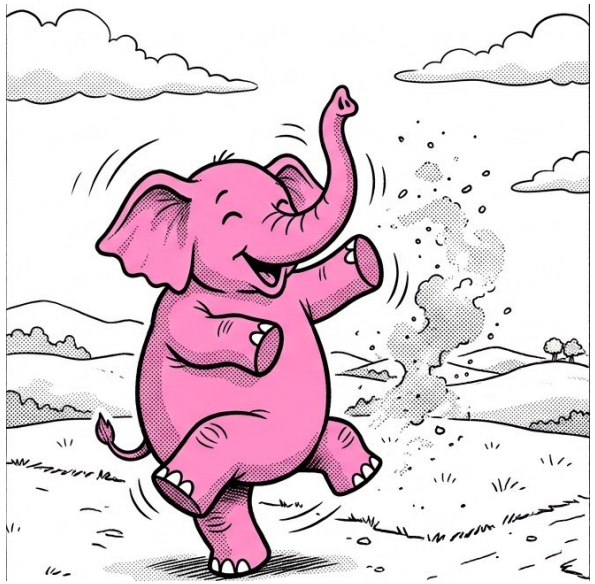



Fun off-label uses thus far:

- Contrasting where “manually-gated” cells end up vs. our unsupervised methods (normalization, clustering, etc.)
- Added benefit of looking-at-the-forest: Easy to identify badly cryopreserved specimens where cell populations were lost

# To unlock the comprehensive biological discovery potential, we need to address these pink elephants





- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?



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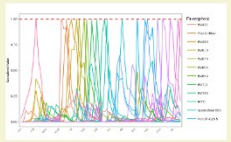
# Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



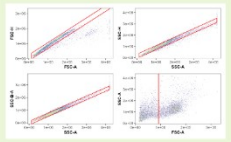

2) Unmixing Controls
(Luciernaga)



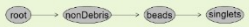
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



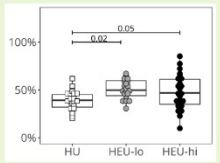
5) Clean Up Gates
(FlowJo or flowWorkspace)




6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)




8) Major subset Outputs

Iterating by Experiment



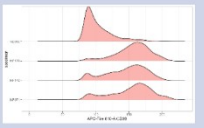

9) Subset Export



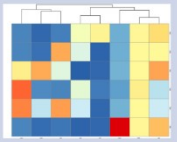
10) Return to R
(flowWorkspace)



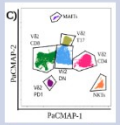

11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)




Summarized Experiment




15) Coereba

Iterating by Cell Subset



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# What good is any tool if the community is not able to use it?

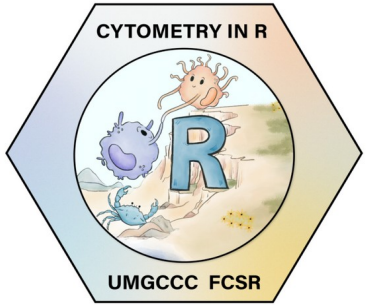
- Few resources to learn Cytometry in R to begin with
- Those that exist aimed at those with intermediate-advanced R skills, not beginners
- Know-how concentrated at a few institutions
- Most self-learning attempts end in frustration

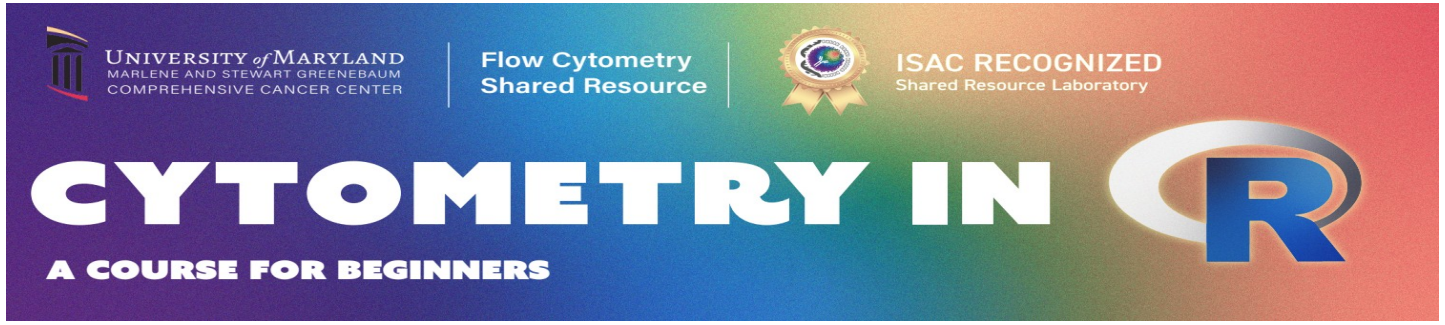




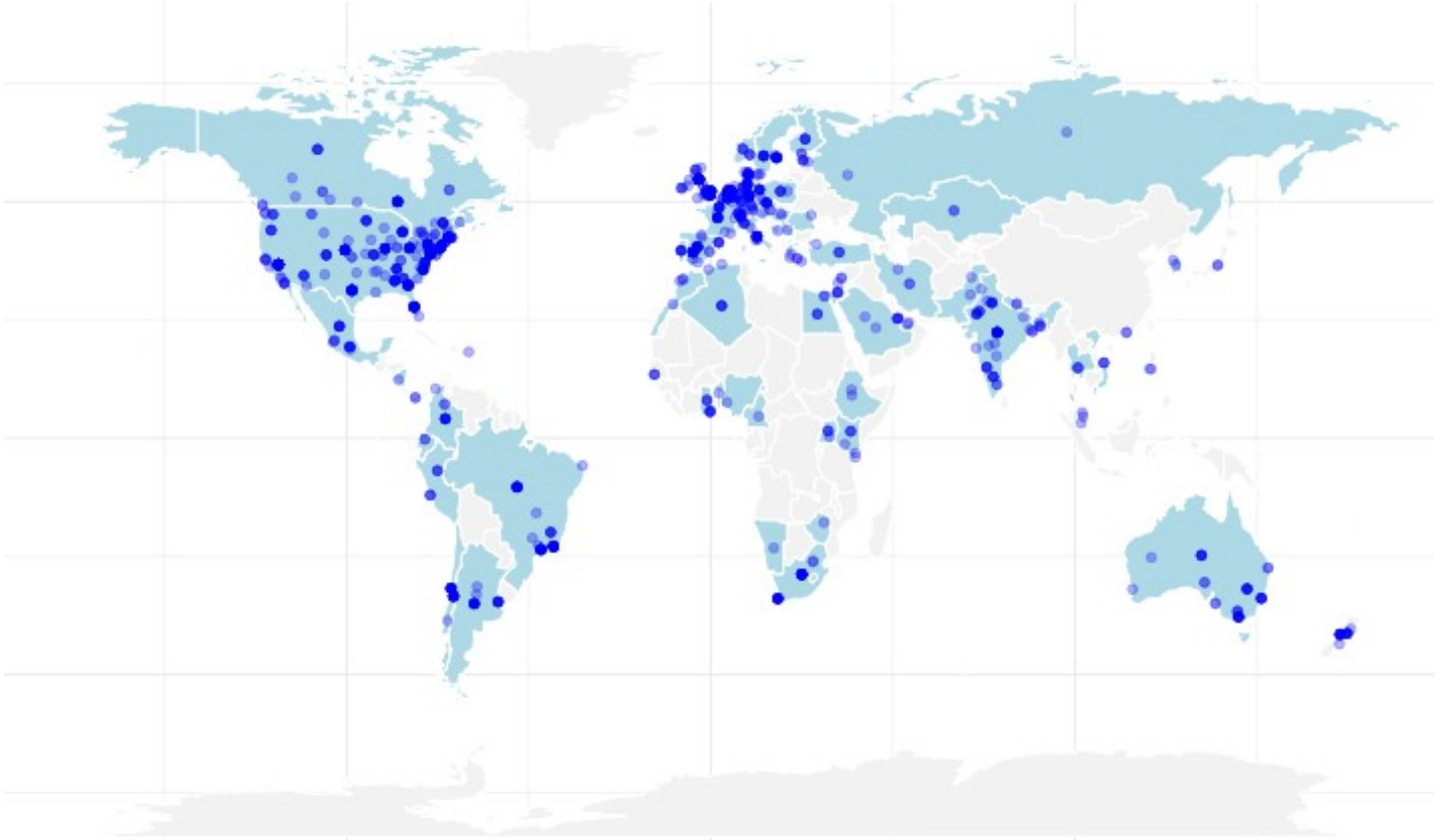
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# Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience



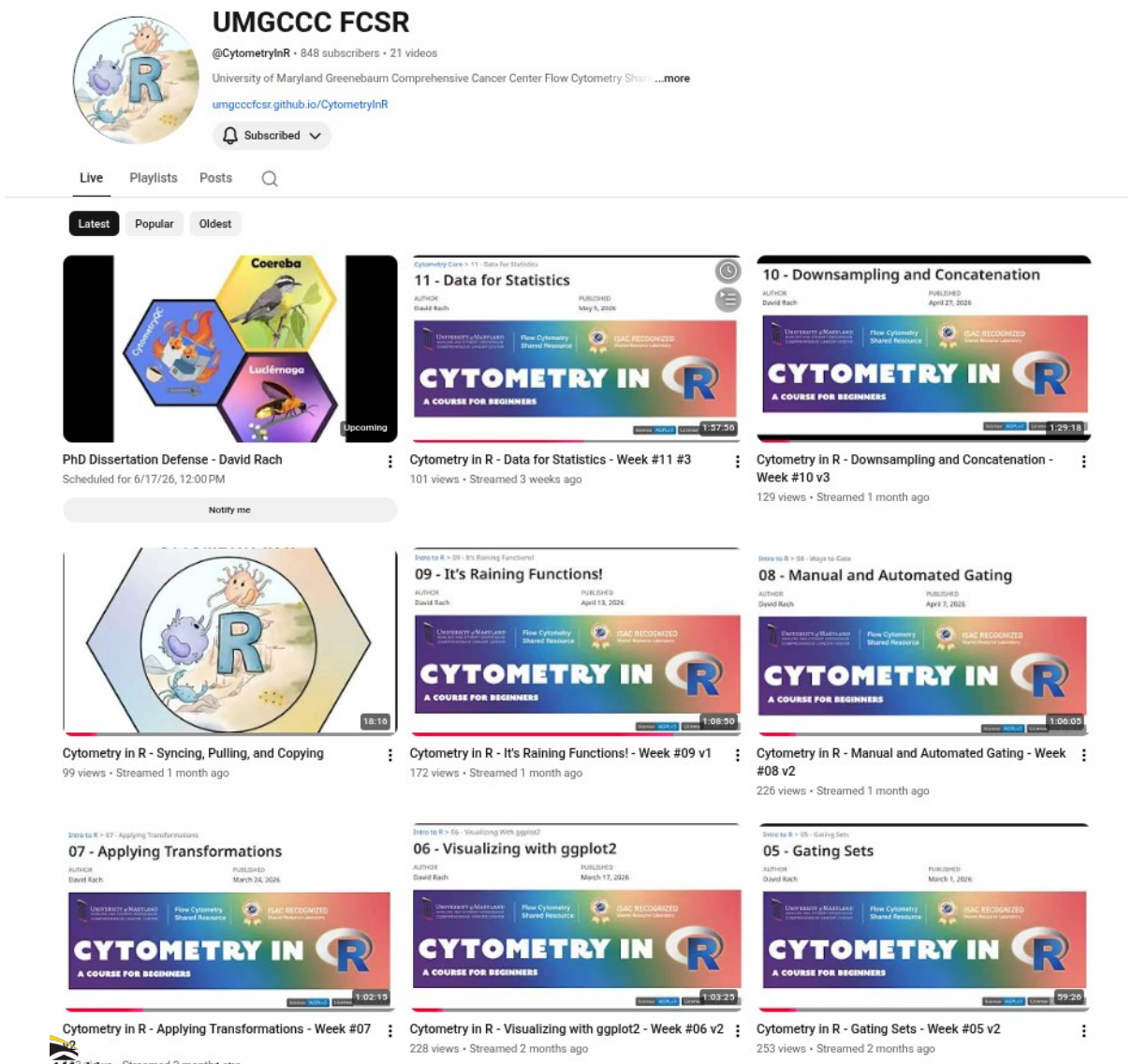

Website



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# YouTube



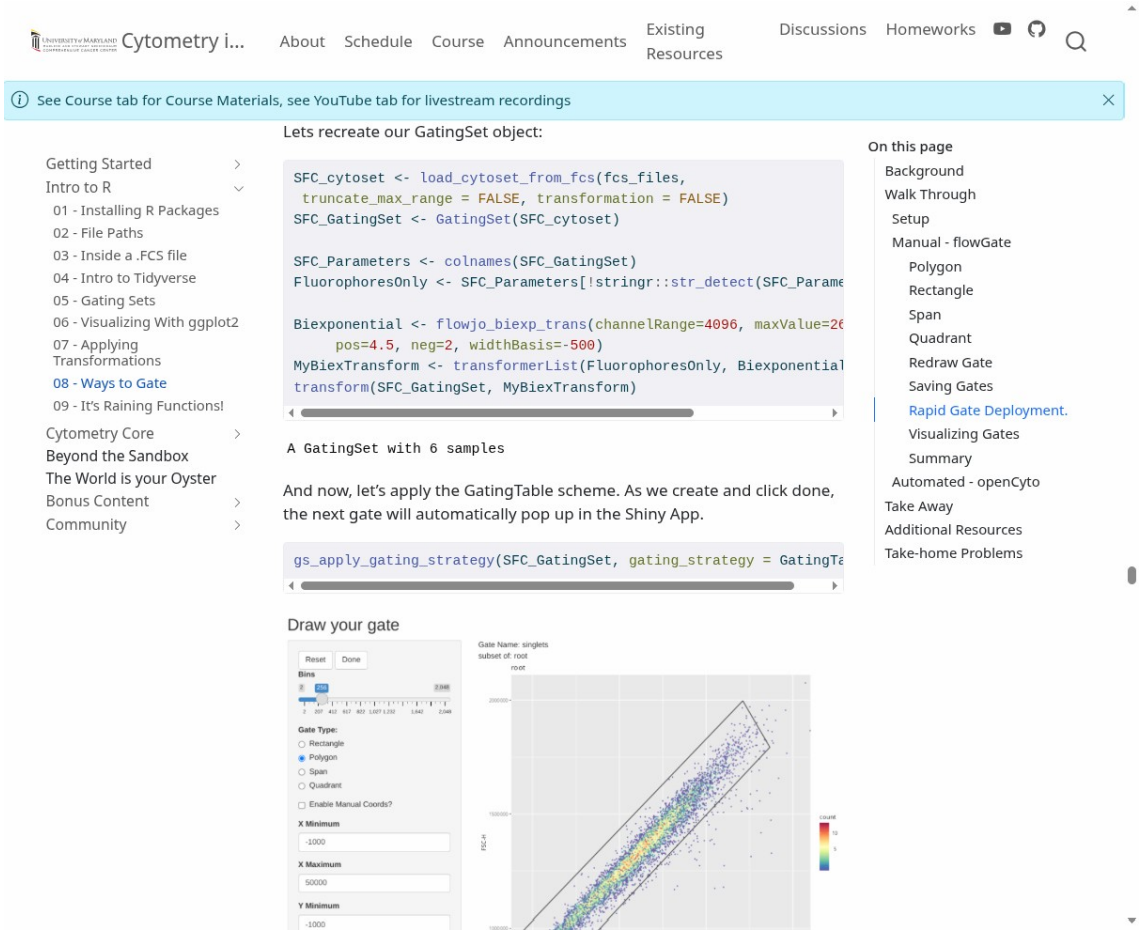

- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing



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# Website

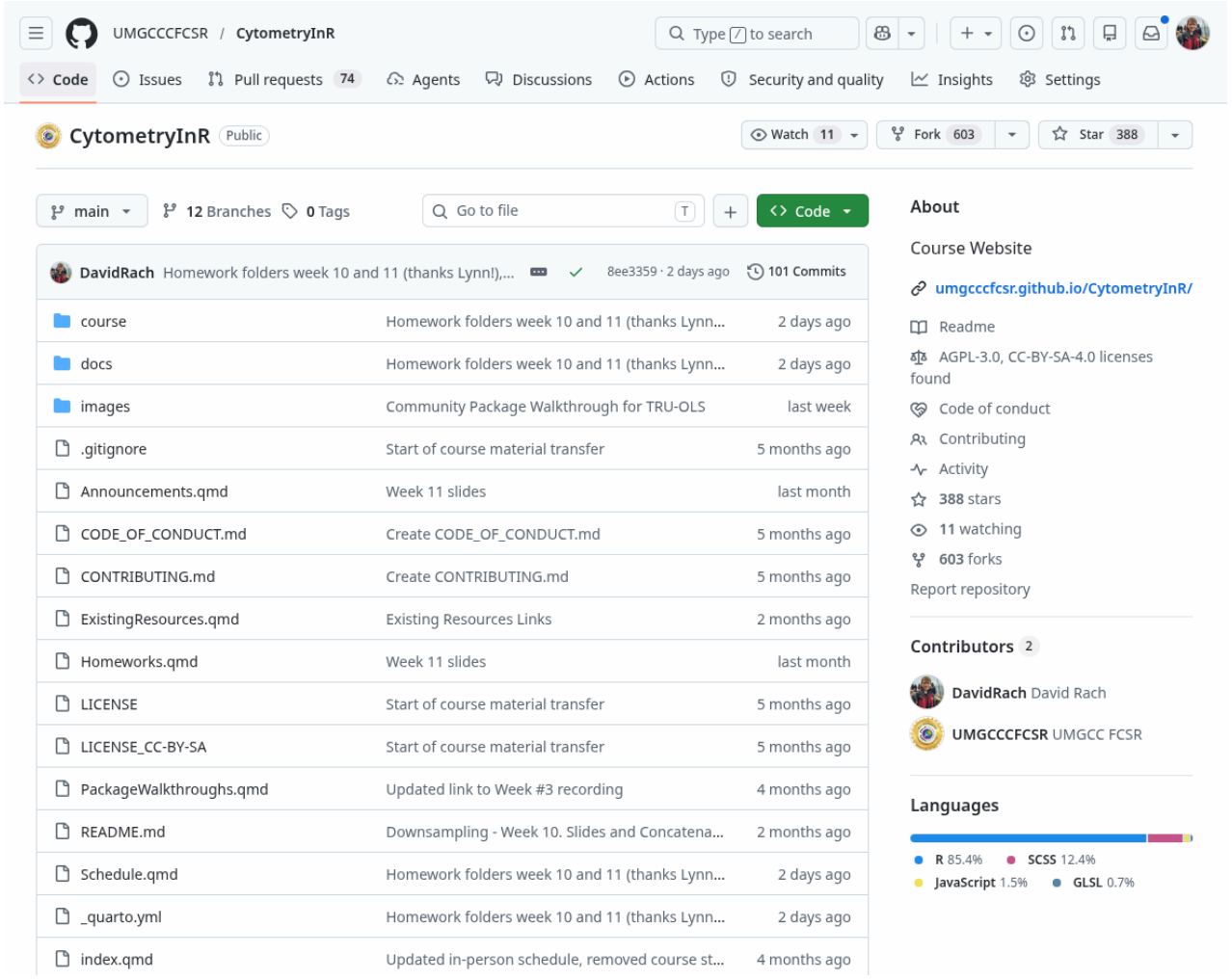
- Created with Quarto, hosted via GitHub pages.
- Code, code outputs, screenshots and explanations
- Searchable
- New weekly material added as a new .qmd file






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# GitHub

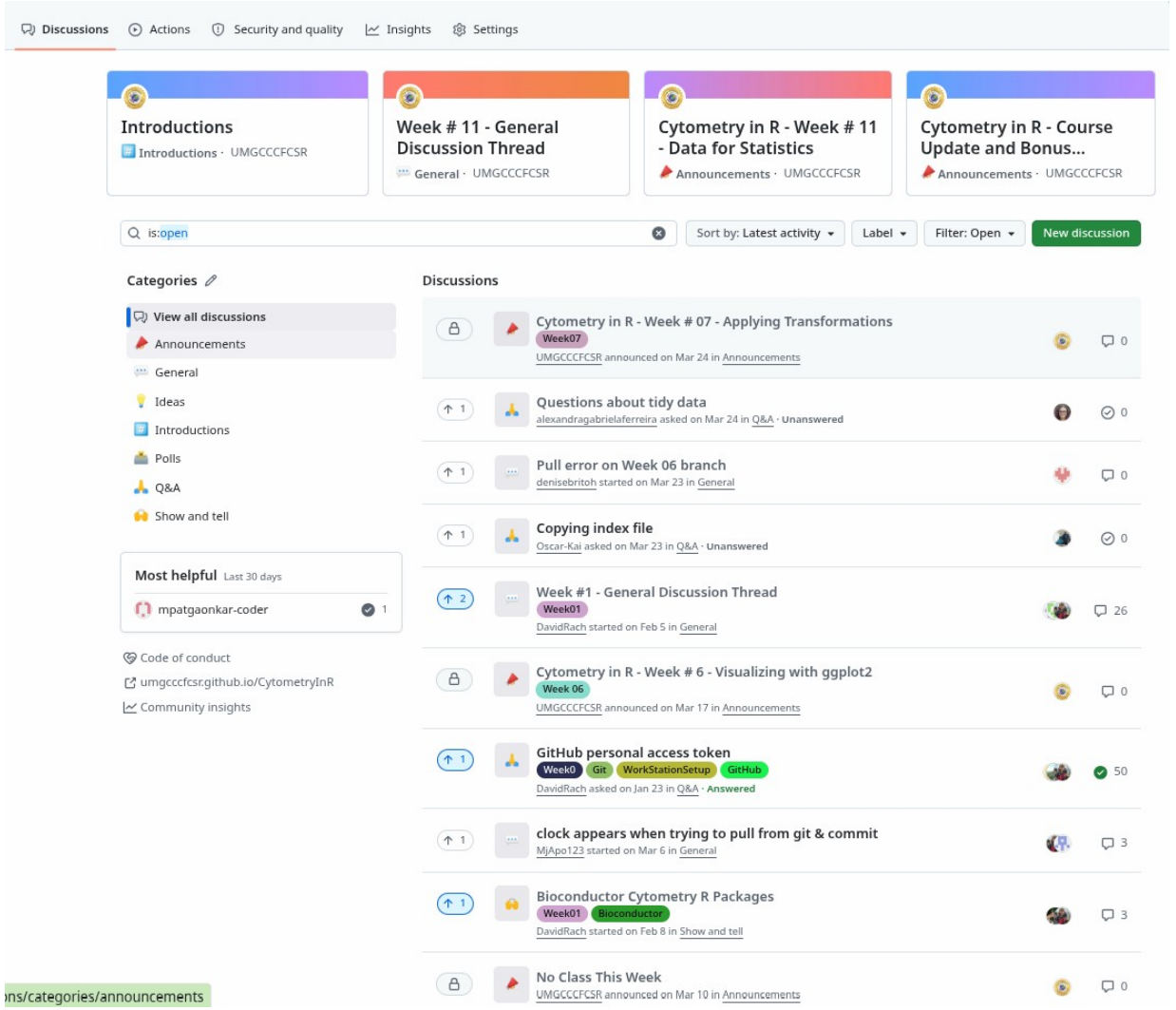



- Code and datasets available via the GitHub repository
- Course participants start course by creating GitHub account and forking the repository
- After Positron is linked to GitHub account, pull locally
- When new material posted each week, sync and pull

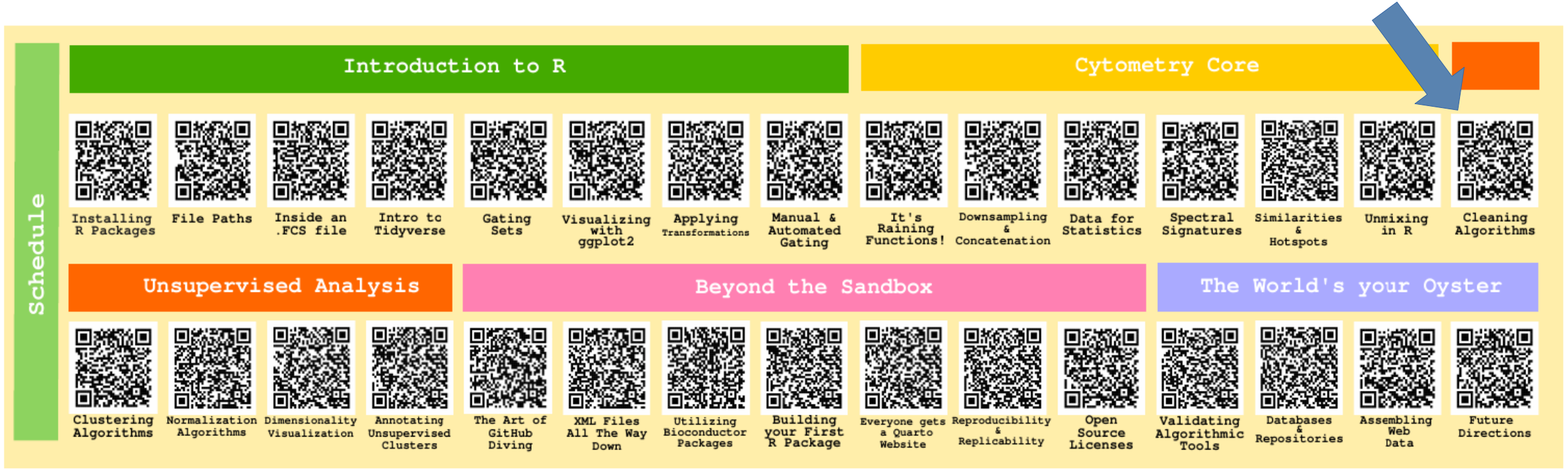
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# Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.



# Half-way through this passion project

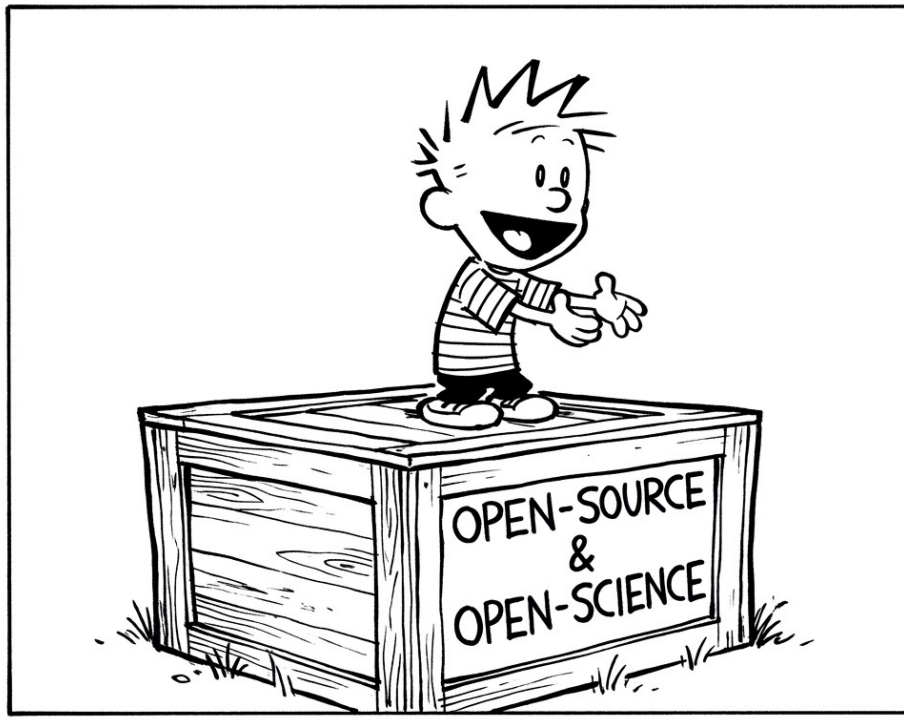



- Wrapping up Cytometry Core module
- Gathering steam to start on the Unsupervised Analysis section



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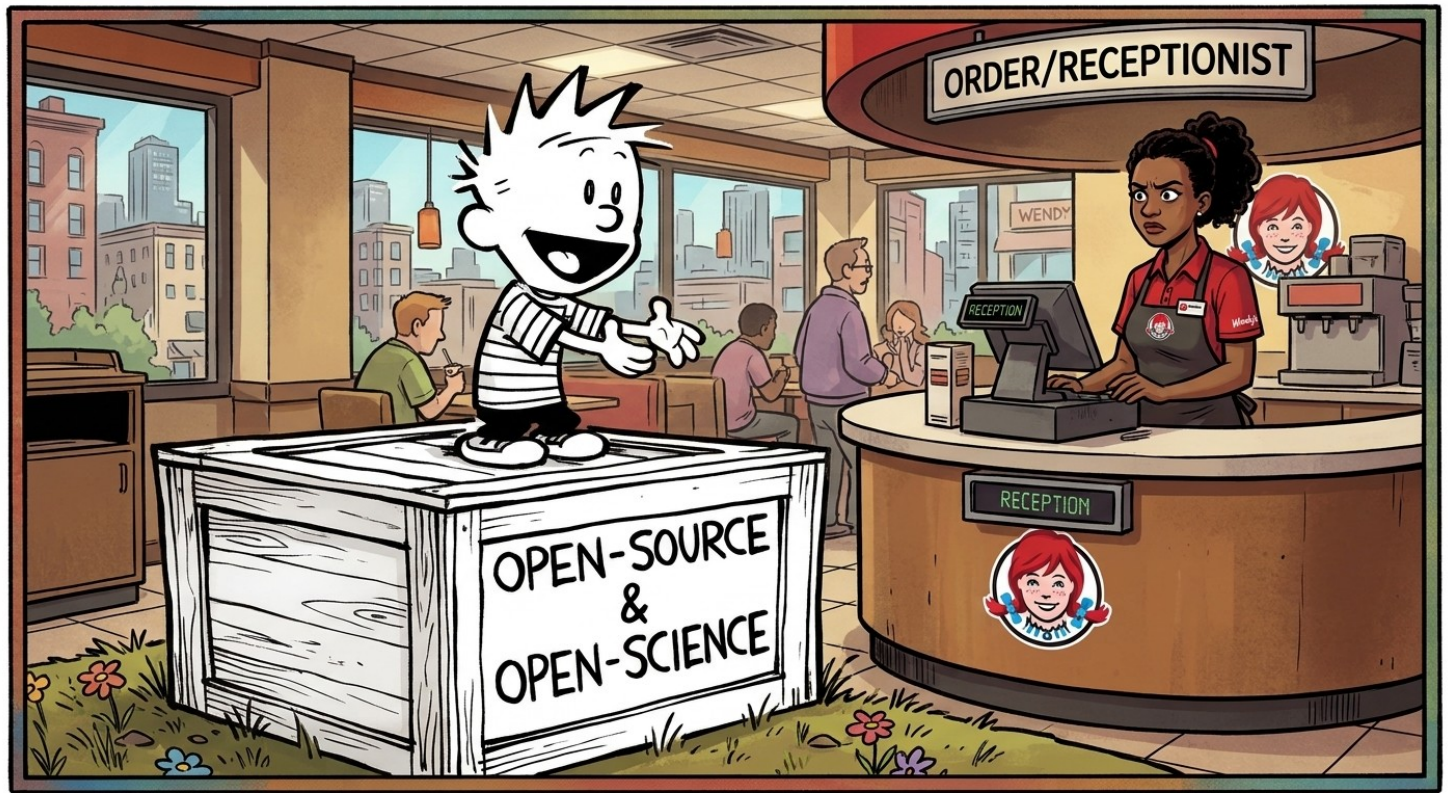
# As scientist, our community is our strength






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# Just my two over-caffeinated cents






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# Acknowledgments

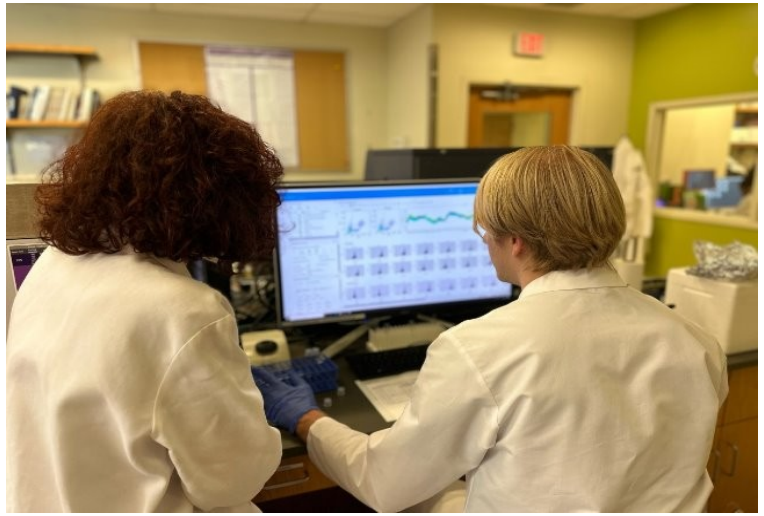
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OF HEALTH SCIENCES

All mothers and their infants, and their communities

All the open-source contributors, developers, and maintainers, that collectively made the tool sets without which all of this would not be feasible

Dr. Cristiana Cairo
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Dr. Xiaoxuan Fan

Dr. Daniela Franco
Dr. Natarajan Ayithan



U01 HD092309

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Cytometry Discord

Bioconductor Zulip

https://github.com/DavidRach

@davidrach.bsky.social

www.linkedin.com/in/david-rach/

Want to collaborate? Reach out!

Hiring? 👀

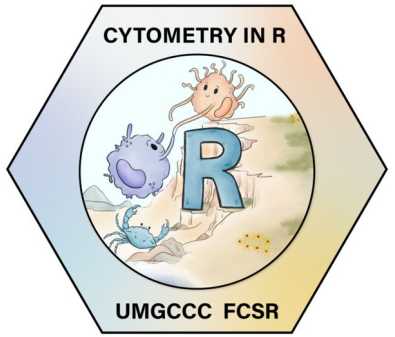
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Shared Resource Laboratory

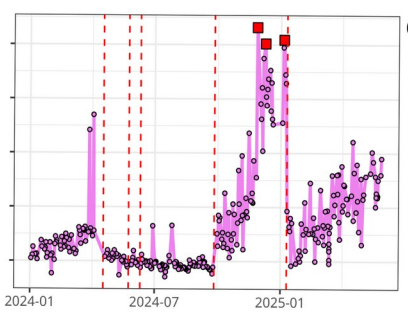
# Questions?




Luciernaga



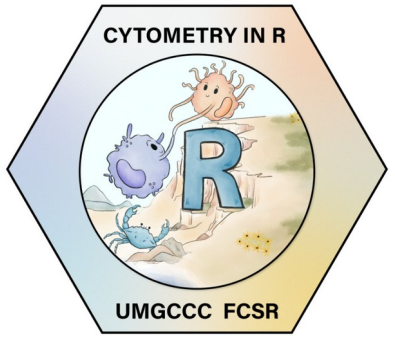

Website - CytometryInR




UMGCCC FCSR
InstrumentQC dashboard




Coereba




YouTube @**CytometryInR**

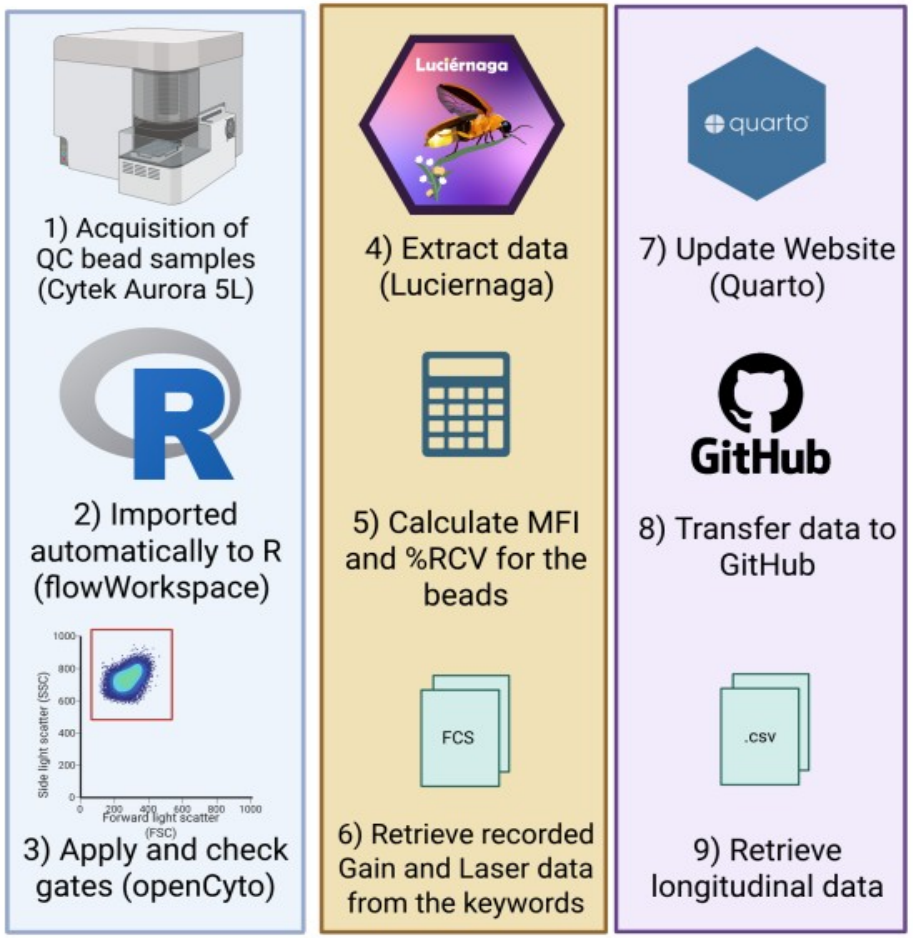



InstrumentQC How-To



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# Automated Daily Processing







Behind-the-scenes:

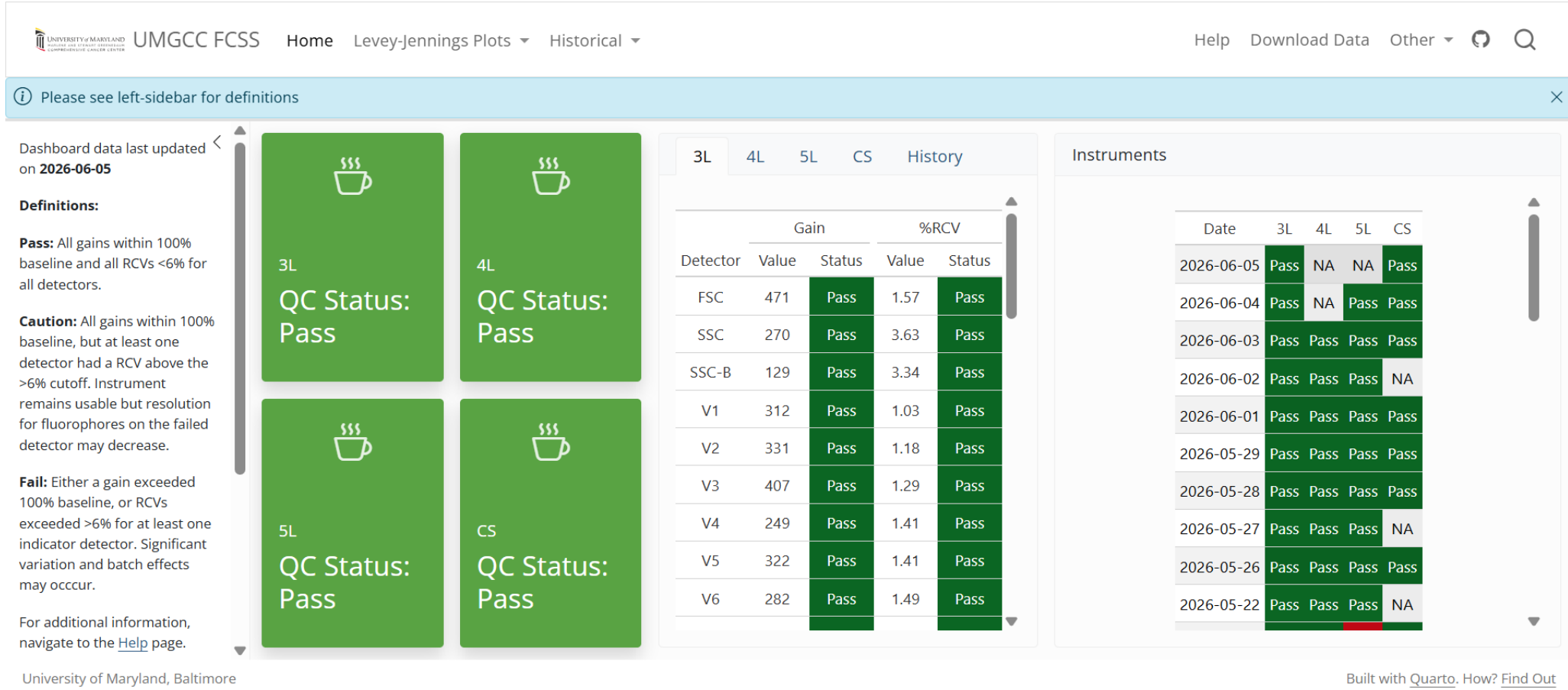
- Using R (and increasingly Rust) to process the .fcs files
- Automated transfer to GitHub via Windows Task Scheduler
- Our Quarto Website gets updated twice daily via a GitHub action

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# Instrument Quality Control Dashboard

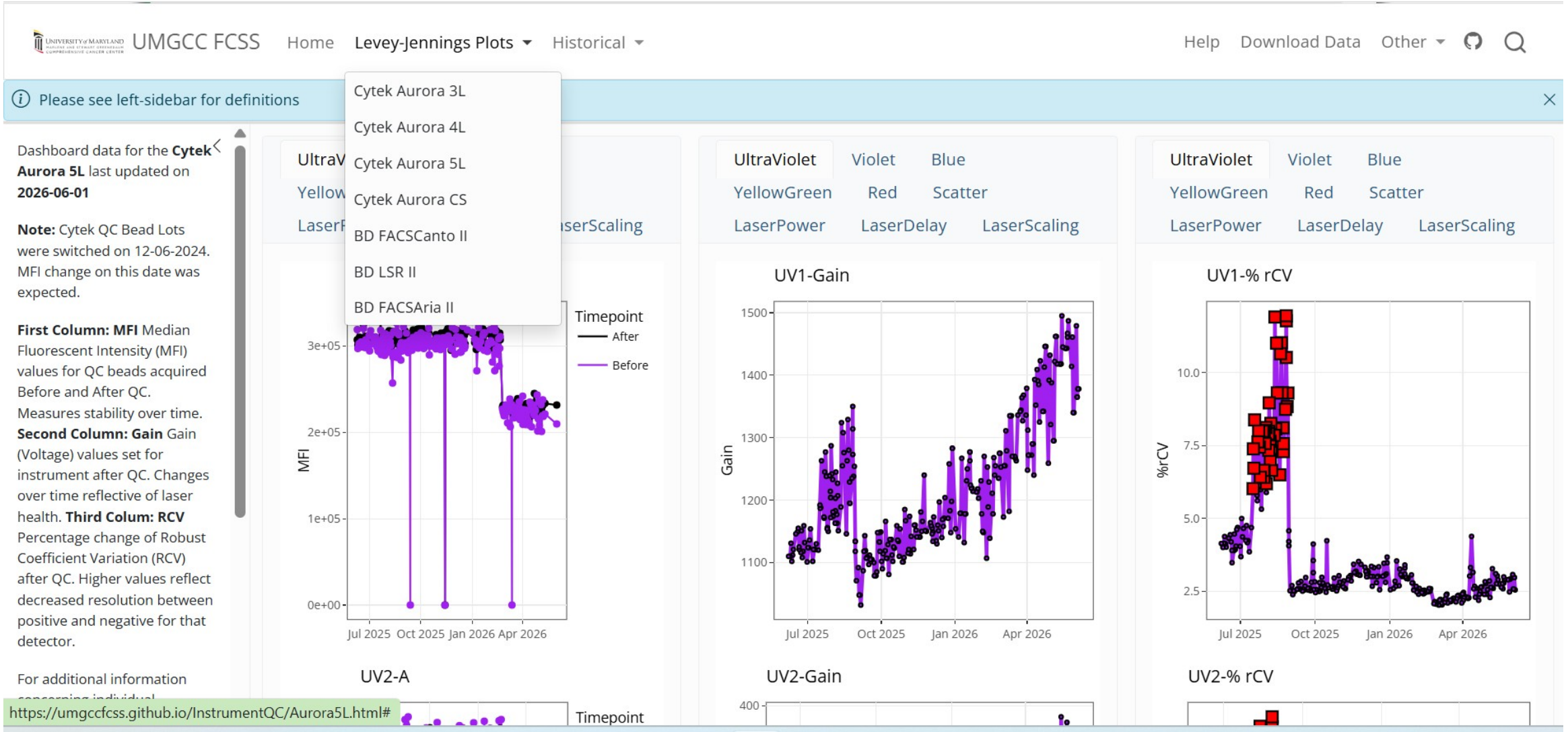
Dashboard

How-To




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# Instrument Quality Control Dashboard




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